SLOVENSKI STANDARD
SIST EN 13053:2020

01-februar-2020

Nadomešča:

SIST EN 13053:2007+A1:2011

Prezračevanje stavb - Klimatske naprave - Ocenjevanje in lastnosti naprav, sestavnih delov in sklopov

Ventilation for buildings - Air handling units - Rating and performance for units, components and sections

Lüftung von Gebäuden - Zentrale raumlufttechnische Geräte - Leistungskenndaten für Geräte, Komponenten und Baueinheiten

Ventilation des bâtiments - Caissons de traitement d'air - Classification et performance des unités, composants et sections

Ta slovenski standard je istoveten z: EN 13053:2019

ICS:

91.140.30	Prezračevalni in klimatski sistemi	Ventilation and air-conditioning systems

SIST EN 13053:2020 **en,fr,de**

2003-01.Slovenski inštitut za standardizacijo. Razmnoževanje celote ali delov tega standarda ni dovoljeno.

SIST EN 13053:2020

EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 13053

December 2019

ICS 91.140.30

Supersedes EN 13053:2006+A1:2011

English Version

Ventilation for buildings - Air handling units - Rating and performance for units, components and sections

Ventilation des bâtiments - Centrales de traitement d'air - Classification et performance des unités, composants et sections

Lüftung von Gebäuden - Zentrale raumlufttechnische Geräte - Leistungskenndaten für Geräte, Komponenten und Baueinheiten

This European Standard was approved by CEN on 28 September 2018.

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Ref. No. EN 13053:2019 E

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European foreword

This document (EN 13053:2019) has been prepared by Technical Committee CEN/TC 156 "Ventilation for buildings", the secretariat of which is held by BSI.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by June 2020, and conflicting national standards shall be withdrawn at the latest by June 2020.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. CEN shall not be held responsible for identifying any or all such patent rights.

This document supersedes EN 13053:2006+A1:2011.

This document has been prepared under a standardization request given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive.

For relationship with EU Directive, see informative Annex ZA, which is an integral part of this document.

This document has been revised and includes new requirements according to Ecodesign requirements for ventilation units given in EU Commission Regulation No 1253/2014.

According to the CEN-CENELEC Internal Regulations, the national standards organisations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Republic of North Macedonia, Romania, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey and the United Kingdom.

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Introduction

This European Standard is a part of a suite of standards for air handling units used for ventilation and air conditioning of buildings for human occupancy. It considers the ratings and the performance of air handling units as a whole, the requirements and performance of specific components and sections of air handling units including hygiene requirements. The position of this standard in the field of mechanical building services is shown in Figure 1.

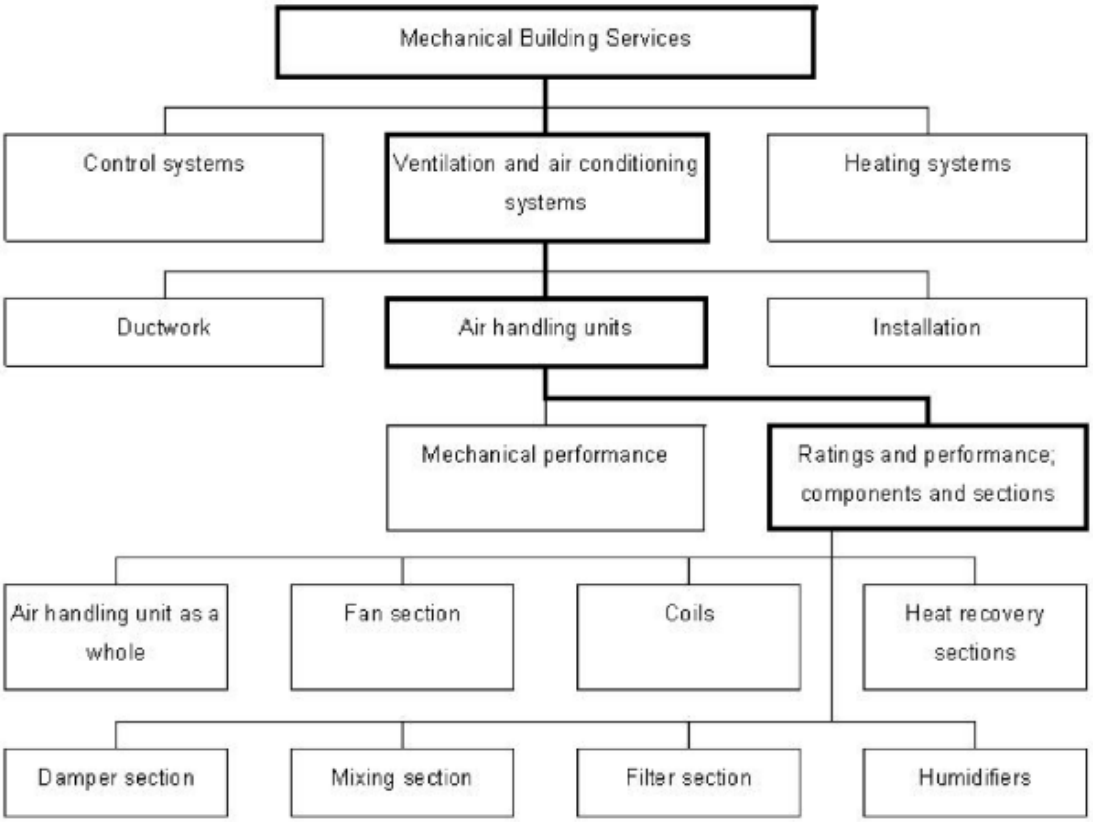

Figure 1 — Position of this standard in the field of mechanical building services

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1 Scope

This document specifies requirements and testing for rating and performance of Non Residential Ventilation Units, NRVU's, specifically Air Handling Units (AHU's). It specifies requirements, classifications and testing of components and sections of air handling units.

This document applies to tests in a laboratory and *in situ*. This document is applicable both for mass produced air handling units and tailor-made Air Handling Units.

This document applies to AHU and individual sections of AHU with the designed air flow > 250 $m^3 \cdot h^{-1}$. This document applies to UVU's with additional air treatment components in addition to filtration.

This standard does not include:

— residential unidirectional and bidirectional ventilation units;

— nonresidential unidirectional ventilation units which consist of only a casing, a fan with or without filter.

NOTE 1 Residential ventilation units are covered by EN 13142.

NOTE 2 Nonresidential unidirectional ventilation units which consists only casing, fan with or without filter are covered by EN 17291.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 308, *Heat exchangers — Test procedures for establishing performance of air to air and flue gases heat recovery devices*

EN 1216, *Heat exchangers — Forced circulation air-cooling and air-heating coils — Test procedures for establishing the performance*

EN 1751, *Ventilation for buildings — Air terminal devices — Aerodynamic testing of damper and valves*

EN 1886, *Ventilation for buildings — Air handling units — Mechanical performance*

EN 12599:2012, *Ventilation for buildings — Test procedures and measurement methods to hand over air conditioning and ventilation systems*

EN 12792:2003, *Ventilation for buildings — Symbols, terminology and graphical symbols*

EN 16211, *Ventilation for buildings. Measurement of air flows on site. Methods*

EN ISO 3741, *Acoustics — Determination of sound power levels and sound energy levels of noise sources using sound pressure — Precision methods for reverberation test rooms (ISO 3741)*

EN ISO 3743-1, *Acoustics — Determination of sound power levels and sound energy levels of noise sources using sound pressure — Engineering methods for small movable sources in reverberant fields — Part 1: Comparison method for a hard-walled test room (ISO 3743-1)*

EN ISO 3744, *Acoustics — Determination of sound power levels and sound energy levels of noise sources using sound pressure — Engineering methods for an essentially free field over a reflecting plane (ISO 3744)*

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EN ISO 3746, *Acoustics — Determination of sound power levels and sound energy levels of noise sources using sound pressure — Survey method using an enveloping measurement surface over a reflecting plane (ISO 3746)*

EN ISO 5136, *Acoustics — Determination of sound power radiated into a duct by fans and other air-moving devices — In-duct method (ISO 5136)*

EN ISO 5167-1, *Measurement of fluid flow by means of pressure differential devices inserted in circular cross-section conduits running full — Part 1: General principles and requirements (ISO 5167-1)*

EN ISO 5801, *Industrial fans — Performance testing using standardized airways (ISO 5801)*

EN ISO 7235, *Acoustics — Laboratory measurement procedures for ducted silencers and airterminal units — Insertion loss, flow noise and total pressure loss (ISO 7235)*

EN ISO 9614-1, *Acoustics — Determination of sound power levels of noise sources using sound intensity — Part 1: Measurement at discrete points (ISO 9614-1)*

EN ISO 9614-2, *Acoustics — Determination of sound power levels of noise sources using sound intensity — Part 2: Measurement by scanning (ISO 9614-2)*

EN ISO 9614-3, *Acoustics — Determination of sound power levels of noise sources using sound intensity — Part 3: Precision method for measurement by scanning (ISO 9614-3)*

EN ISO 16890-1, *Air filters for general ventilation — Part 1: Technical specifications, requirements and classification system based upon particulate matter efficiency (ePM) (ISO 16890-1)*

EN ISO 16890-2, *Air filters for general ventilation — Part 2: Measurement of fractional efficiency and air flow resistance (ISO 16890-2)*

EN ISO 16890-3, *Air filters for general ventilation — Part 3: Determination of the gravimetric efficiency and the air flow resistance versus the mass of test dust captured (ISO 16890-3)*

EN ISO 16890-4, *Air filters for general ventilation — Part 4: Conditioning method to determine the minimum fractional test efficiency (ISO 16890-4)*

ISO 3966, *Measurement of fluid flow in closed conduits — Velocity area method using Pitot static tubes*

ISO 13347-1, *Industrial fans — Determination of fan sound power levels under standardized laboratory conditions - Part 1: General overview*

ISO 13347-2, *Industrial fans — Determination of fan sound power levels under standardized laboratory conditions - Part 2: Reverberant room method*

ISO 13347-3, *Industrial fans — Determination of fan sound power levels under standardized laboratory conditions - Part 3: Enveloping surface methods*

ISO 13347-4, *Industrial fans — Determination of fan sound power levels under standardized laboratory conditions - Part 4: Sound intensity method*

ISO 16956, *Thermal performance in the built environment — Determination of air flow rate in building applications by field measuring methods*

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3 Terms and definitions

For the purposes of this document, the terms and definitions given in EN 12792:2003 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

— IEC Electropedia: available at http://www.electropedia.org/

— ISO Online browsing platform: available at http://www.iso.org/obp

3.1
air handling unit
factory made unit encased assembly containing a fan or fans, filtrating device(s) and other necessary equipment to perform at least one or more of the following functions: heating, cooling, heat recovery, heat transfer, humidifying, dehumidifying and mixing air

3.2
section
< air handling unit > functional element of an air handling unit consisting of one or more components in a single casing

3.3
component
< air handling unit > smallest functional element of an air handling unit

3.4
casing
< air handling unit > enclosure of the unit, within which the components are mounted

3.5
damper section
section of air handling unit including a damper or valve

3.6
mixing section
section where e.g. outdoor air flow and the recirculation air flow are mixed in a controlled way

Note 1 to entry: The section generally consists of one damper per air flow and a mixing chamber

3.7
filter section
section including a filter or filters and an associated filterframe

3.8
heat recovery section
section in which heat (and possibly also moisture) is transferred from one airstream into another, either directly or using an intermediary heat transfer medium

3.9
heating and cooling coils
heat exchangers by means of which heat is transferred from a heat transfer medium to air (heating coil) or the other way round (cooling coil)

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3.10
sound attenuation section
section in which sound transfer into a ductwork or into ambient air is reduced

3.11
humidifier section
section in which moisture is added to the air

3.12
fan section
section in which one or more fans are installed for moving air

3.13
functions

3.13.1
air treatment
process by which the state of the air is modified with respect to one or more of its characteristics such as temperature, moisture content, dust content, bacterial count, gas and vapour content

3.13.2
cooling
removal of latent and/or sensible heat

3.13.3
dehumidification
controlled reduction of water vapour from the air

3.13.4
filtration
removal of particulate material from the airstream

3.13.5
heating
transfer of heat from one body or medium to another medium

3.13.6
humidification
controlled addition of water vapour to an air stream or space

3.13.7
sound attenuation
controlled reduction of sound energy

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3.14
characteristics

3.14.1
air flow
movement of air within set boundaries (such as ducts)

3.14.2
air flow rate
volume flow of air passing a given plane divided by time

3.14.3
bypass leakage
unwanted and uncontrolled passing of untreated air into the treated air between the components within a casing, such as filters and coils

3.14.4
defrosting heat factor
ratio between the energy transferred into the air supply and the maximum recoverable energy in exhaust air, excluding the energy input for defrosting

3.14.5
unit pressure, p_u
difference between the total pressure at the outlet of the air handling unit and the total pressure at the inlet

3.14.6
unit static pressure p_{us}
difference between the static pressure at the outlet of the air handling unit and the total pressure at the inlet

3.14.7
external static pressure difference, $\Delta p_{s,ext}$
difference between the static pressure at the outlet of the air handling unit and the static pressure at the inlet

3.14.8
humidification efficiency
ratio between the mass of water evaporated by the humidifier and the theoretical mass needed to achieve saturation at a given temperature

3.15
mass produced air handling unit
air handling units manufactured to the same design and by the same manufacturing process, with a range of performance characteristics for which the duty point is not defined

Note 1 to entry: Mass produced air handling units (AHUs) are designed for an undefined duty point and mostly produced in large quantities. Their capacity is generally given in ranges so that they can be used in different buildings and/or applications.

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3.16
tailor made air handling unit
air handling unit individually designed and manufactured for a specific duty point, with defined and known parameters, designed on a specific order and for a specific purpose

3.17
nominal external pressure
declared design external static pressure difference at nominal air flow

3.18
internal pressure difference of ventilation components, $\Delta p_{s,int}$
sum of the static pressure differences of a reference configuration of a BVU or an UVU at nominal air flow

3.19
internal pressure difference of additional non-ventilation components, $\Delta p_{s,add}$
remainder of the sum of all internal static pressure differences at nominal air flow after subtraction of the internal pressure difference of ventilation components (Δps,int)

3.20
internal specific fan power of ventilation components SFP_{int}
ratio between the internal pressure difference of ventilation components and the fan efficiency, determined for the reference configuration

3.21
nominal duty point
point at declared design flow rate of an air handling unit at standard air conditions and at declared external pressure which is used to assess SFP_{int} and thermal efficiency

3.22
nominal air flow
declared flow rate at the nominal duty point

3.23
reference configuration of a NRVU-BVU
product configured with a casing, at least two fans with variable speed or multi-speed drives, a HRS, a clean fine filter (minimum ISO ePM1 50%) on the supply air side and a clean medium filter (minimum ISO ePM10 50%) on the exhaust air side

3.24
reference configuration of a NRVU-UVU supply air
product configured with a casing and at least one fan with variable speed or multi-speed drive, and in case the product is intended to be equipped with a filter, this filter shall be a clean fine filter (minimum ISO ePM1 50%)

3.25
reference configuration of a NRVU-UVU exhaust air
product configured with a casing and at least one fan with variable speed or multi-speed drive, and in case the product is equipped with a filter, no filter class is specified

4 Symbols and abbreviations

For the purposes of this standard, symbols and units given in EN 12792:2003 and in Table 1 apply together with those defined by the formulas, text and annexes of this standard.

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Table 1 — Symbols, terms, units and subscripts

Symbol	Term	Unit
c	Sound velocity in the air	$m \cdot s^{-1}$
E	Duct end correction value	dB
f	Frequency	Hz
h	Enthalpy	
n	Number of measurements within the total measuring time	-
L_p	Sound pressure level	dB
L_W	Sound power level	dB
L_{WA}	A-weighted sound power level	dB(A)
n_{fan}	Rotational speed of the fan	s^{-1}
P_{el}	Electrical motor input power, including any speed control	W
$P_{el,aux}$	Auxiliary electric power input (e.g. pumps)	W
$P_{el,int}$	Electrical power consumption due to internal pressure differences of ventilation components	W
Pm_{ref}	Reference power input	W
p_a	Atmospheric pressure	Pa
p_d	Dynamic pressure	Pa
p_s	Static pressure	Pa
p_t	Total pressure	Pa
p_u	Unit pressure	Pa
p_{us}	Unit static pressure	Pa
q_d	Mass flow of water drain and overflow	$kg \cdot s^{-1}$
q_m	Air mass flow rate	$kg \cdot s^{-1}$
q_v	Air volume flow rate	$m^3 \cdot s^{-1}$
q_{vm}	Measured and converted air volume flow rate	$m^3 \cdot s^{-1}$
q_{vs}	Specified air volume flow	$m^3 \cdot s^{-1}$
q_w	Mass flow of water inlet	$kg \cdot s^{-1}$
S	Section area	m^2
SFP_{int}	Specific Fan Power, internal	$W \cdot m^{-3} \cdot s$
u	Uncertainty range of measured data	%
U	range of uniformity of flow	
v	Velocity of air at a point	$m \cdot s^{-1}$
x	Absolute humidity	$g \cdot kg^{-1}$
$\Delta p_{s,ext}$	External static pressure difference	Pa

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$\Delta p_{s,fan}$	Static pressure difference of the fan	Pa
$\Delta p_{s,int}$	Internal pressure difference of ventilation components (reference configuration)	Pa
ε	Coefficient of performance	-
ε_D	Defrosting heat factor	-
η_D	Average overall static efficiency of power consumption	-
η_e	Energy efficiency	-
η_{ea}	Yearly energy efficiency	-
$\eta_{fs,fan}$	Overall static efficiency including motor and drive efficiency of the individual fan(s) in the ventilation unit determined at an air flow and an external pressure	-
η_h	Humidifier efficiency	-
η_{mix}	Mixing efficiency	%
η_t	Temperature efficiency	
ρ	Density	$kg \cdot m^{-3}$
ρ_{st}	Standard air density	$kg \cdot m^{-3}$
φ	Relative humidity	%
Ω	Solid angle of radiation at the end of the duct	rad
Subscripts		
1	Inlet	
2	Outlet	
add	Related to additional pressure difference of non-ventilation components	
cas	Related to AHU casing	
corr	Corrected value	
EHA	Related to exhaust air	
ETA	Related to extract air	
el	Related to electric input	
ext	External	
fan	Related to the fan	
F	Related to clean filter	
HRS	Heat Recovery System	
int	Internal	
H	Air flow with higher temperature	
L	Air flow with lower temperature	
M	Mixed air flow [mean temperature]	

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meas	Measured value
max	Maximum value
min	Minimum value
s	Related to static pressure
s,ext	Related to external static pressure difference of the unit
s,free,fan	Related to static pressure of the stand-alone fan
SUP	Related to supply air
TE	Related to test conditions
tot	Air flow downstream of the mixing section
u	Related to the air handling unit (not to the fan)
Abbreviations	
AHU	Air Handling Unit
HVAC	Heating, ventilation and air conditioning
HRS	Heat Recovery System
BVU	Bidirectional Ventilation Unit
UVU	Unidirectional Ventilation Unit
NRVU	Non Residential Ventilation Unit

5 Ratings and performance of the entire air handling unit

5.1 General

The performance of the entire air handling unit cannot be defined as the sum of the individual components and sections. Hence, the procedures that follow shall be applied to a complete air handling unit. In particular cases, under agreed circumstances, and in order to define the value of SFP_{int}, these procedures can be applied to a part of an air handling unit.

The methods described in 5.2 cover among other things measuring air volume flow together with pressures and power consumption. By selecting an appropriate test system, these procedures can be extended to include measuring the sound level transmitted from the air handling unit into the ductwork at a known volume flow, as described in 5.3.

5.2 Testing of aerodynamic performance

5.2.1 Test method

5.2.1.1 Basis of method

Tests shall be carried out in accordance with one of the methods shown in EN ISO 5801. The most suitable test installation type B, C or D shall be adopted considering the on-site or intended installation, the geometry of the air-handling unit and the facilities available.

The three installation types are as follows:

a) installation Type B: free inlet, ducted outlet;

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b) installation Type C: ducted inlet, free outlet and

c) installation Type D: ducted inlet, ducted outlet.

In the above classification, the terms shall be taken to have the following meanings:

Free inlet or outlet signifies that air enters or leaves the air handling unit directly from or to the unobstructed free atmosphere. Ducted inlet or outlet signifies that air enters or leaves the unit through a duct directly connected to the unit inlet or outlet.

5.2.1.2 Chamber test method

Where a standardized test chamber is used, it shall conform to the requirements of EN ISO 5801.

5.2.1.3 Ducted test method

The common parts of a ducted system, for Types B, C or D installations, shall conform to the requirements of EN ISO 5801. The cross-sectional dimensions of the air outlet shall be used to determine the dimensions of the outlet ducting required in a Type B or Type D installation, and the inlet ducting required in a Type C or Type D installation.

5.2.2 Characteristics and quantities

5.2.2.1 Characteristics

a) Air flow rate/pressure performance

For air handling units installed or intended to be installed in accordance with installation category B and D: Relationship between unit pressure p_u (as relevant design pressure parameter) and air volume flow rate q_v.

For air handling units installed or intended to be installed in accordance with installation category C: Relationship between unit static pressure p_{us} (as relevant design pressure parameter) and air volume flow rate q_v.

b) Electrical power

Relationship between electrical input power and air volume flow rate.

If a speed adjustment device is included, e.g. frequency inverter, the electrical motor input power shall include the power of speed control devices.

These characteristics shall be converted from the ambient temperature and pressure measured at the time of the test, to standard conditions with an air density of 1,2 kg × m^{-3}. These characteristics shall be presented for a stated nominal fan speed but without adjustment for inherent speed deviation during the test.

5.2.2.2 Quantities

a) *Air volume flow rate* (q_v) shall be measured by any method which is in accordance with EN ISO 5801, ISO 16956, EN 12599, EN 16211, EN ISO 5167-1 or ISO 3966, e.g. a nozzle, an orifice plate or a pitot-static tube.

b) *Unit pressure* (p_u) and *Unit static pressure* (p_{us}) shall be determined from the pressure measurements defined in 5.2.3.2. The dynamic pressure as part of the total pressure is related to the respective measurement plane. The duct sizes shall be the sizes defined by the manufacturer.

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NOTE The simplified calculation method for units supplying a pressure lower than 2000 Pa with a Mach number of less than 0,15 according to ISO 5801 can be applied.

Unit pressure p_u:

$$p_u = p_{t2} - p_{t1} = p_{s2} - p_{s1} + p_{d2} - p_{d1} \quad (1)$$

Unit static pressure p_{us}:

$$p_{us} = p_{s2} - p_{t1} = p_{s2} - p_{s1} - p_{d1} \quad (2)$$

where

p_{t2} is the sum of the static pressure and the dynamic pressure at unit outlet, expressed in Pa;

p_{s2} is the static pressure at unit outlet, expressed in Pa.

p_{d2} is the dynamic pressure at unit outlet, expressed in Pa.

p_{t1} is the sum of the static pressure and the dynamic pressure at unit inlet, expressed in Pa.

p_{s1} is the static pressure at unit inlet, expressed in Pa.

p_{d1} is the dynamic pressure at unit inlet, expressed in Pa.

c) *Density of air (ρ)* shall be given in kg · m^{-3}, by using Formula (3) according to EN ISO 5801:

$$\rho = \frac{p_a - 0{,}378 p_v}{287\left(273 + t_a\right)} \quad (3)$$

where

p_a is the atmospheric pressure, expressed in Pa;

p_v is the partial pressure of water vapour in the air, expressed in Pa;

287 is the gas constant of dry air, expressed in J · kg^{-1} · K^{-1};

t_a is the dry-bulb temperature, expressed in °C.

d) *Temperature of the air (t_a)* shall be measured at the point of flow measurement.

e) *Rotational speed of the fan (n_{fan})* shall be measured at each test point.

f) *Electrical motor input power (P_{el})*, the power to the fan motor, shall be measured at each test point. The applied voltage and the current to each phase shall also be recorded when measured.

g) *Static pressure difference of the fan $\Delta p_{s,fan}$*, expressed in Pa, shall be determined at nominal duty point.

5.2.2.3 Determination of *external static pressure difference $\Delta p_{s,ext}$*

The relevant external pressure size to be considered in the determination of the eco-design parameters SFP_{int} and fan efficiency is the *external static pressure difference $\Delta p_{s,ext}$*. It can be measured either directly considering measurement requirements described in 5.2.3.2 or it can be derived from the air flow rate/pressure characteristics.

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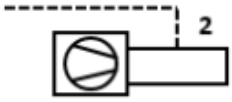
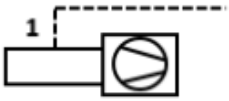
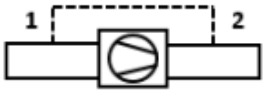
The correlation between $\Delta p_{s,ext}$ and the air flow rate/pressure characteristics p_u (unit pressure) and p_{us} (unit static pressure) of the air handling unit is shown in Figure 2.

	Installation type B	Installation type C	Installation type D
$\Delta p_{s,ext}$			
direct measurement	p_{s2}	$-p_{s1}$	$p_{s2} - p_{s1}$
calculated based on unit static pressure p_{us}	p_{us}	$p_{us} + \frac{\rho_{st}}{2} \cdot v_1^2$	$p_{us} + \frac{\rho_{st}}{2} \cdot v_1^2$
calculated based on unit pressure p_u	$p_u - \frac{\rho_{st}}{2} \cdot v_2^2$	—	$p_u - \frac{\rho_{st}}{2} \cdot (v_2^2 - v_1^2)$

Key

p_{s1} is the static pressure at inlet side (pressure difference against ambient pressure, Pa);

p_{s2} is the static pressure at outlet side (pressure difference against ambient pressure, Pa);

ρ_{st} is the density of 1,2 kg · m^{-3} corresponding to the air under standard conditions (20 °C, 101 325 Pa);

v is the calculated average velocity in the duct at the respective measurement plane (in m · s^{-1})

Figure 2 — Determination of external static pressure difference

5.2.3 Measurement procedure

5.2.3.1 Conditions for measurement

5.2.3.1.1 General

Dampers that control the flow of air in the part of the air handling unit to be tested shall be fully open. Other dampers that form part of a different air circuit, e.g. bypass and recirculation dampers, shall be fully closed.

All elements included in the design of the air handling unit shall be fitted as intended with filters (average of the measured initial and defined final pressure loss at designed airflow – see 6.9.2) and dry coils. If there is no negative influence on the internal pressure of the unit, the average filter pressure difference shall be simulated by increasing the external total pressure difference of the unit with a value equal to the difference between rated average and initial filter pressure difference.

To determine the unit performance with a loaded filter an extra pressure drop shall be added to simulate the loaded filter during testing.

The aerodynamic performance of mass produced air handling units shall be measured with clean filters. The filter clogging shall be considered by the designer when specifying the duty point.

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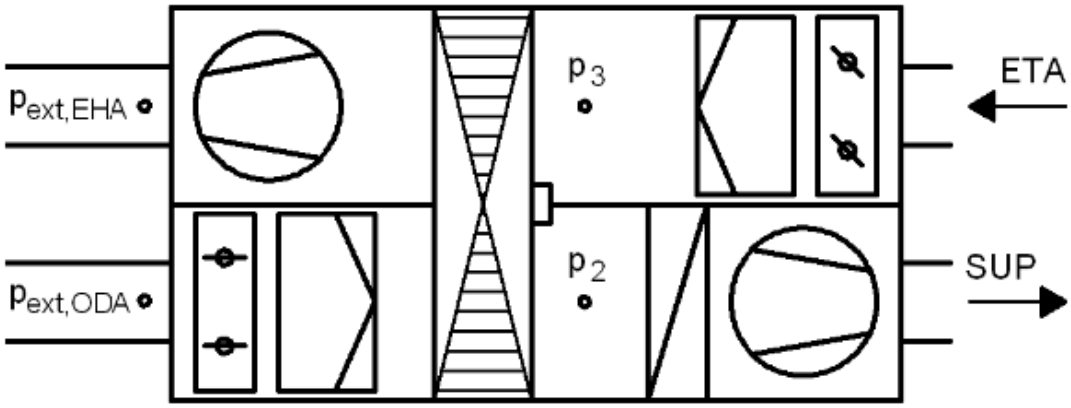
5.2.3.1.2 Testing of unit with heat recovery

5.2.3.1.2.1 General

Testing shall be performed taking the leakage between the air streams into consideration.

5.2.3.1.2.2 Testing of complete unit (both air streams)



Key

EHA	exhaust air
ODA	outdoor air
ETA	extract air
SUP	supply air

Figure 3 — Testing of complete unit

The airflow shall be measured at the supply air side and at the extract air side. The external pressure shall be set to design pressure conditions. Unless otherwise stated, the pressure difference on the outdoor airside and exhaust airside is set to 50 Pa. The remainder of the external pressures shall be set on the supply and extract air openings. In order to avoid leakages from the extract air stream to the supply air stream, the pressure p_2 should be higher than the pressure p_3. The two pressures p_2 and p_3 shall be measured. The leakage and the extra pressure difference are the responsibility of the manufacturer.

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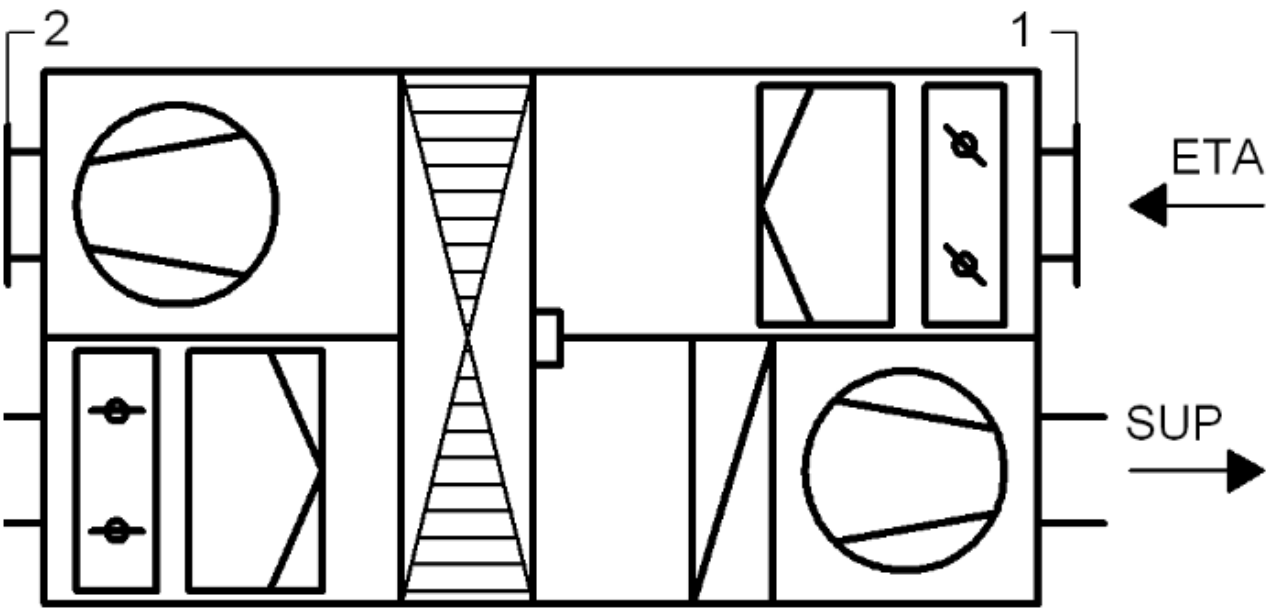
5.2.3.1.2.3 Testing of one air stream



Key

1	inlet plate
2	outlet plate
ETA	extract air
SUP	supply air

Figure 4 — Testing of one air stream

If just one air stream is to be tested, then the connections of the opposite air stream shall be closed with airtight plates.

5.2.3.2 Measurements

Atmospheric pressure and temperature shall be measured during the test. Pressure measurements, at the locations and in the manner described in EN ISO 5801, shall be recorded at a sufficient number of test points enabling the characteristic curve to be plotted through the specified duty point or over the full duty range, whichever is required.

Rotational speed of the fan and the electrical input to the fan motor shall be recorded at each of the test points.

5.3 Testing of acoustic performance

5.3.1 General

5.3.1.1 Acoustic tests

5.3.1.1.1 Test methods

Test shall be done on the range 100 Hz – 10 000 Hz in one-third octave band, or 125 Hz – 8 000 Hz in octave bands.

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Methods using a diffuse field environment (reverberation room) are given in ISO 13347-2, EN ISO 3741, EN ISO 3743-1, EN ISO 3743-2.

Methods using a free field environment are given in ISO 13347-3, ISO 13347-4, EN ISO 3744, EN ISO 3745, EN ISO 9614-1 to –3.

Methods using in duct environment are given in ISO 5136.

For those laboratory tests, the grade of accuracy of the acoustic standard shall be at least grade 2 "engineering method"such as: EN ISO 3743-1, EN ISO 3743-2, EN ISO 3744, EN ISO 3746 (if it meets the engineering method criterions), EN ISO 9614-1, EN ISO 9614-2, ISO 13347-2 or grade 1 "precision method" such as EN ISO 3741, EN ISO 3745, EN ISO 9614-3. Grade 3 methods "survey" such as EN ISO 3746 shall be limited to on-site measurements or factory acceptance tests.

5.3.1.1.2 Case radiated noise test

The casing radiated noise emitted by the complete air handling unit shall be determined in accordance with the test method described in 5.3.1.1.1 and test installation described in 5.3.2.1.

NOTE In the case of air handling units with free inlets or outlets, the casing radiated sound level includes the sound emitted by the free inlet or outlet.

5.3.1.1.3 Duct borne noise tests

Measurement of the sound levels transmitted by the unit into the inlet ducting and the outlet ducting shall be conducted in accordance with the test method described in 5.3.1.1.1 and test installation described in 5.3.2.2

5.3.1.2 Duty point

The air handling unit shall work at the duty point defined by the air handling unit manufacturer.

5.3.1.3 Ductwork

The ductworks shall be sized to match the manufacturer's recommended outlet or inlet opening and shall maintain a constant cross section. Ductwork lengths shall be at least 3 equivalent duct diameters, but not less than 2,6 m. The duct can have both circular and rectangular cross section, according to the section of the unit.

When testing in accordance with EN ISO 5136 appropriate transitions shall be made so that testing can be carried out according to the standard.

5.3.2 Specific requirements concerning the set-up of acoustic tests

5.3.2.1 Casing radiated noise tests

5.3.2.1.1 Test set-up

The measurement of the sound power level emitted by the air openings and casing of the unit shall be performed using one of the test set-ups shown in Figure 5.

Figures 5 a) and 5 b) show the test set-ups for measurement using a diffuse field environment.

Figure 5 c) shows the test set-up for measurement using the free field environment.

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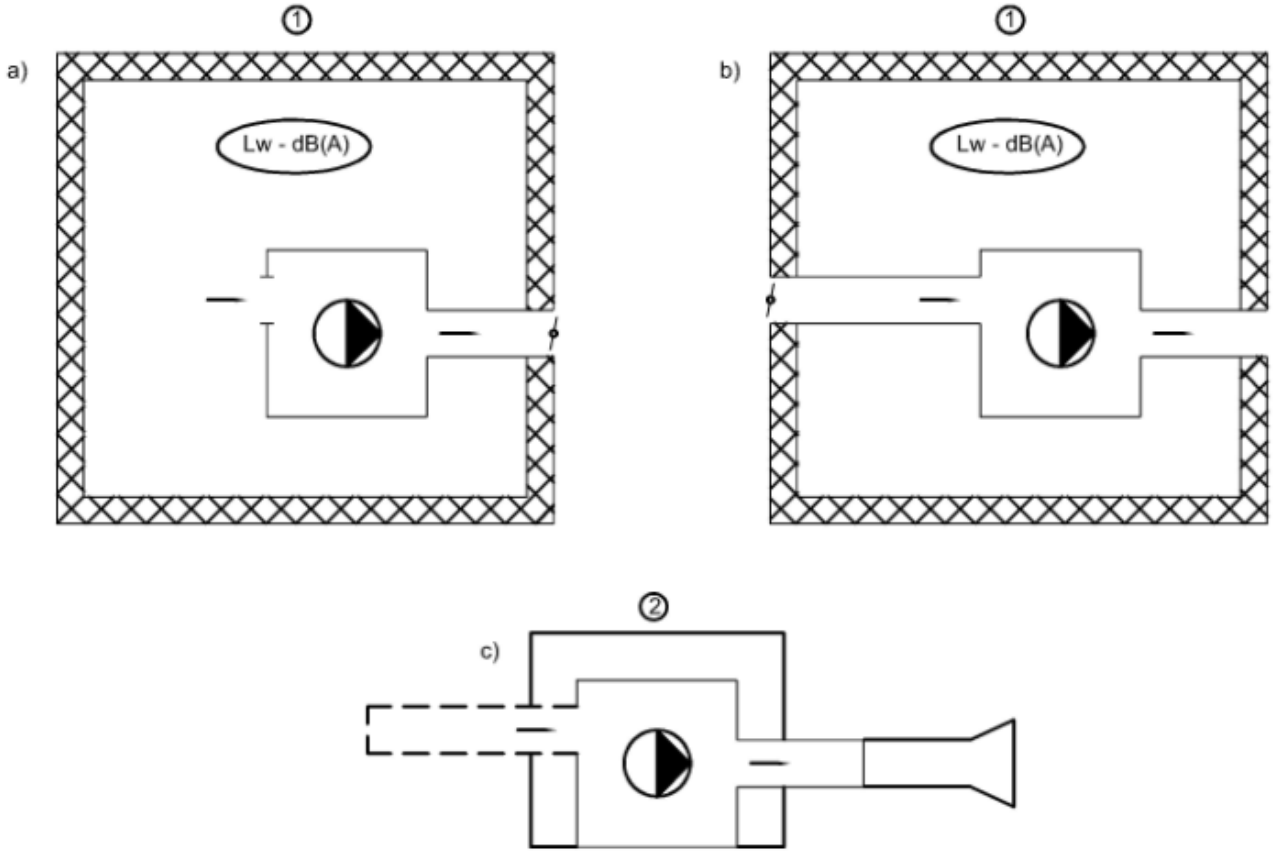

Key

1 reverberation room

2 measuring surface

NOTE Figure 5 a): Set up for units with one free in-/outlet.

NOTE Figure 5 c): Inlet duct is optional.

Figure 5 — Measurement of airborne noise emitted by the air openings and the casing of the unit

5.3.2.1.2 Noise emitted from the ductwork

For test set-ups according to Figures 5 a) but specially Figures 5 b), the construction of the ducts is crucial. The ducts shall be of high transmission loss construction to avoid sound radiating from the ducting, contributing to the airborne noise casing radiated measurements. Confirmation tests shall be conducted to verify that the acoustic contribution from the ductwork is insignificant. For example, successive layers of a low absorption acoustical barrier shall be added to the exterior of the ductwork until the resulting sound measurement indicates no change greater than 1 dB on octave bands from the previous sound measurement in the band of interest.

NOTE Circular ducts have a better sound insulation than rectangular, especially in the low and medium frequency range.

5.3.2.1.3 Throttling device

If a throttling device is necessary for adjusting the unit to the duty point it shall be placed far away from the casing or outside the room in order to avoid its contribution to the resulting sound power level.

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5.3.2.2 Duct borne noise tests

5.3.2.2.1 Test set-up

The measurement of the sound power level transmitted by the unit into the ductwork shall be performed using one of the test set-ups shown in Figure 6.

Figure 6 a) shows the set-up for the measurement using a diffuse field environment.

Figure 6 b) shows the set-up for the measurement using the free field environment. The duct shall end flush with the wall or a baffle.

Figure 6 c) shows the set-up for measuring using anechoic termination. This measurement shall be performed in accordance with EN ISO 5136.

Except for measurement done into the duct with EN ISO 5136 (Figure 6 c)), the duct correction shall be applied in accordance with 5.3.2.2.4.

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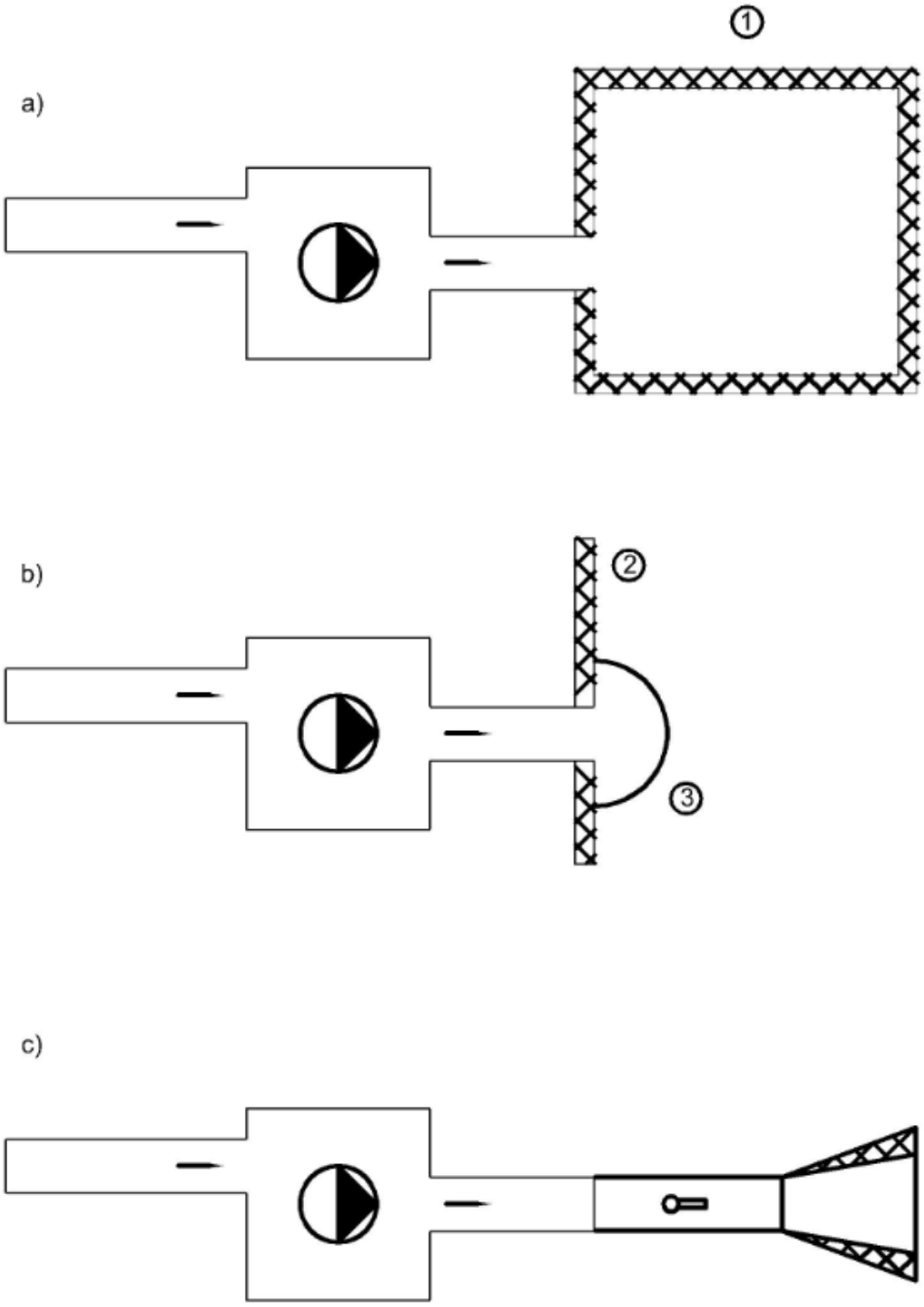

Key

1 reverberation room

2 baffle

3 measuring surface

Figure 6 — Measurement of noise transmitted by the unit into the ductwork

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5.3.2.2.2 Throttling device

Where a throttling device is necessary for adjusting the unit to the duty point, it shall be positioned so that the sound pressure level generated in the test duct by the throttling device is at least 10 dB below the sound pressure level in the test duct from the unit. It is recommended to install an attenuator to reduce the influence of the noise emitted by the throttling device. Throttling device shall not be positioned in the duct where the measurement is performed.

5.3.2.2.3 Baffle

When using free field environment methods a baffle shall be used to simulate a reflecting plane (see Figure 6 b)). The baffle shall be larger than the enveloping surface of measurement and it shall be large enough to provide a barrier for the sound emitted by the unit. This baffle, used to simulate the wall, shall be made of a reflecting plate with sound absorption less than 0,1 with a minimum size of 1 m in each direction. Example: a duct with section of 0,3 m × 0,4 m will require a plate of 2,3 m × 2,4 m, the duct end shall be in the centre of the plate.

5.3.2.2.4 Duct end correction

The end reflection occurs whenever sound is transmitted across an abrupt change in an area such as at the end of a duct in a room or in a free space. When end reflection occurs, some of the sound is reflected back into the duct and is not transmitted into the room or space in which sound measurement is done For this reason, a duct end correction shall be applied to the sound power level measured. The calculation of the duct end correction E depends on the section and geometry of the duct end.

$$E = 10\lg\left[1 + \left(\frac{c}{4\pi f}\right)^2 \Omega\right] \tag{4}$$

where

- E is the duct end correction, expressed in decibel (dB);
- f is frequency, expressed in hertz (Hz);
- c is the speed of sound in air, expressed in meters per second ($m \cdot s^{-1}$), re. 344 $m \cdot s^{-1}$ at 20 °C;
- S is the section area of the duct at its end, expressed in square meters (m^2);
- Ω is the solid angle of radiation at the end of the duct.

Ω is equal to 4π for a duct terminating in a free space at a distance from a surface greater than, or equal to L_{HWL}. L_{HWL} here is a distance equal to one half wavelength of the central frequency of the lowest frequency band of interest.

For a duct terminating flush to the wall, $\Omega = 2\pi$.

For a duct terminating flush to a corner of 2 walls, $\Omega = \pi$.

This correction shall be calculated for each frequency band and added to each sound power determined outside the duct to get the spectrum of sound level inside the duct.

5.4 Testing of internal specific fan power of ventilation components

5.4.1 General

5.4.1.1 Methods

There are two methods to determine SFP_{int} resulting from the pressure losses of ventilation components and overall fan static efficiency. SFP_{int} shall be calculated for the reference configuration.

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The first method (internal) is to measure the static pressure difference of the fan and component pressure losses internally to the AHU via static pressure tappings. This method can be used when there is sufficient space in the AHU between components so that accurate pressure measurements can be made.

The second method (external) is to determine the SFP_{int} by measuring the fan characteristics outside the AHU. This method can be used when the internal space in the AHU is confined such that accurate pressure measurements are difficult.

For regulatory purposes the SFP_{int} shall be calculated for the reference configuration unit at the nominal duty point, see 3.17, 3.21 and 3.22.

The basic test set up and measurements, air volume flow, external static pressure difference and effective electric power input of fan drive (including speed control device if existing), shall be in accordance with 5.2.

5.4.1.2 Quantities

Supplementary to 5.2.2.2.

a) *Static pressure differences* (Δp_x) need to be measured. The measurements should be made according to Annex C and EN 12599:2012, E.2. Figure 7 shows the different pressure difference values in an AHU.

b) *Effective electric input power of fan drive* (P_{el}), shall be measured directly by an instrument that indicates the true output (TRMS) and has a wide frequency range. The measurements are recommended to be made according to EN 12599:2012, D.6. Input power shall include speed control device if existing.

c) *Internal specific fan power of ventilation components* (SFP_{int}) shall be calculated as:

For air handling units with one air side:

$$SFP_{int} = \frac{\Delta p_{s,int}}{\eta_{s,fan}} \tag{5}$$

where

SFP_{int} is the internal specific fan power of ventilation components, expressed in watts per cubic metre per second ($W \cdot m^{-3} \cdot s$);

$\Delta p_{s,int}$ is the internal pressure difference of ventilation components, pascals (Pa);

$\eta_{s,fan}$ is the overall static efficiency drive of the built-in fan (system losses considered) at the considered duty point of the unit.

For air handling units with two air sides:

$$SFP_{int} = \frac{\Delta p_{s,int,SUP}}{\eta_{s,fan,SUP}} + \frac{\Delta p_{s,int,EHA}}{\eta_{s,fan,EHA}} \tag{6}$$

where

SFP_{int} is the internal specific fan power of ventilation components, expressed in watts per cubic metre per second ($W \cdot m^{-3} \cdot s$);

$\Delta p_{s,int,SUP}$ is the internal pressure difference of ventilation components in supply air stream, expressed in pascals (Pa);

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$\eta_{s,fan,SUP}$ is the overall static efficiency drive of the built-in fan (system losses considered) in supply air stream at the considered duty point of the unit.

$\Delta p_{s,int,EHA}$ is the internal pressure difference of ventilation components in exhaust air stream, expressed in pascals (Pa);

$\eta_{s,fan,EHA}$ is the overall static efficiency drive of the built-in fan (system losses considered) in exhaust air stream at the considered duty point of the unit.

If supply and/or exhaust air side consists of more than one air stream (e.g. run around coils systems), then the SFP_{int,air_side} shall be calculated for each side by using the following Formula (7):

$$SFP_{int,air_side} = \frac{P_{el,int,1} + P_{el,int,2} + \ldots + P_{el,int,n}}{q_{v,1} + q_{v,2} + \ldots + q_{v,n}} \quad (7)$$

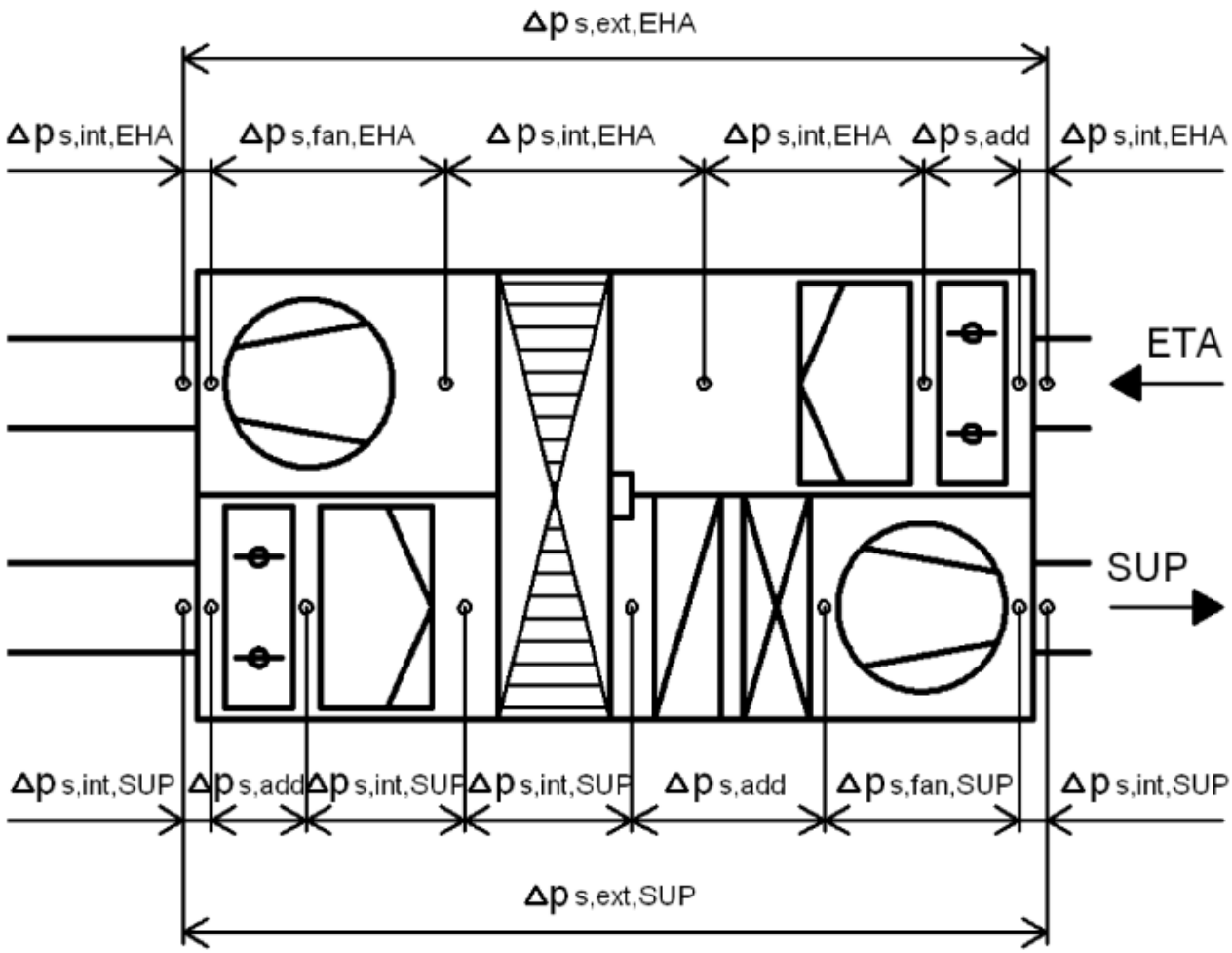

Figure 7 — Air handling unit and pressure definitions

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5.4.1.3 Conditions for measurements

Dampers that control the flow of air in the part of the air handling unit to be tested shall be fully open. Other dampers that form part of a different air circuit, e.g. bypass and recirculation dampers, shall be fully closed.

All elements included in the design of the air handling unit shall be fitted as intended. Filters shall be clean (initial filter pressure difference is requested) and coils dry.

Measurements shall occur at constant nominal airflow rates and constant external pressure.

5.4.1.4 Duty point

Each air stream in the air handling unit shall be set up at its nominal duty point. The respective external static pressure difference shall be determined according 5.2.2.3.

5.4.1.5 Measurements

Atmospheric pressure and temperature shall be measured at the beginning of the test. Additional measurements should be performed if the duration of the test exceeds one hour.

5.4.1.6 Evaluation of results

All static pressure values and effective electric input power of fan drives shall be converted to values corresponding to standard air density of 1,20 kg · m^{-3}.

$$\Delta p_{\mathrm{corr}} = \Delta p_{\mathrm{TE}} \frac{\rho_{\mathrm{st}}}{\rho_{\mathrm{TE}}} \tag{8}$$

$$P_{\mathrm{el}_{\mathrm{corr}}} = P_{\mathrm{el}_{\mathrm{TE}}} \frac{\rho_{\mathrm{st}}}{\rho_{\mathrm{TE}}} \tag{9}$$

where

ρ_{st}	is the standard air density (1,20 kg · m^{-3})
Index: TE	are the measured values under test conditions
Index: corr.	are there calculated values

5.4.2 Test method

5.4.2.1 Testing inside air handling unit

5.4.2.1.1 Measurement procedure

Measure air volume flow through the air handling unit, external static pressure difference, all component pressure differences and effective electric input power of fan drive analogically as in 5.2 for each air stream.

5.4.2.1.2 Built-in fan efficiency

Built-in fan efficiency, $\eta_{\mathrm{s,fan}}$ is the overall static efficiency drive of the built-in fan (system losses considered) at the considered duty point of the unit.

$$\eta_{\mathrm{s,fan}} = \frac{q_{\mathrm{v}} \cdot \Delta p_{\mathrm{s,fan}}}{P_{\mathrm{el}}} \tag{10}$$

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where,

q_v	air flow rate expressed in $m^3 \cdot s^{-1}$
P_{el}	electical power consumption of fan (input power shall include speed control device if existing) expressed in (W)
$\Delta p_{s,fan}$	static pressure difference of the fan expressed in pascals (Pa)

5.4.2.1.3 Calculation of SFP_{int}

The internal specific fan power, for one air stream, results from:

$$SFP_{int} = \frac{P_{el,int}}{q_v} = \frac{\Delta p_{s,int}}{\eta_{s,fan}} \tag{11}$$

where

$$P_{el,int} = \frac{P_{el}}{\Delta p_{s,fan}} \cdot \Delta p_{s,int} \tag{12}$$

where

$$\Delta p_{s,int} = \Delta p_{s,HRS} + \Delta p_{s,F} + \Delta p_{s,cas} = \Delta p_{s,fan} - \Delta p_{s,ext} - \Delta p_{s,add} \tag{13}$$

where

$\eta_{s,fan}$	is the overall static efficiency drive of the built-in fan (system losses considered) at the considered duty point of the unit.
q_v	is the air flow rate expressed in $m^3 \cdot s^{-1}$
P_{el}	is the electical power consumption of fan (input power shall include speed control device if existing) expressed in (W)
$P_{el,int}$	is the electrical power consumption due to internal pressure differences of ventilation components (input power shall include speed control device if existing) expressed in (W)
$\Delta p_{s,int}$	is the internal pressure difference of ventilation components expressed in (Pa)
$\Delta p_{s,HRS}$	is the pressure difference over the heat recovery system expressed in pascals (Pa)
$\Delta p_{s,F}$	is the pressure difference over the clean filter expressed in pascals (Pa)
$\Delta p_{s,cas}$	is the pressure difference over the casing (inlets and outlets) expressed in pascals (Pa)
$\Delta p_{s,fan}$	is the static pressure difference of the fan expressed in pascals (Pa)
$\Delta p_{s,ext}$	are the external pressure losses (duct work) expressed in pascals (Pa)
$\Delta p_{s,add}$	are the additional pressure losses (e.g. cooler, silencer, humidifier) expressed in pascals (Pa).

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5.4.2.2 Testing outside air handling unit

5.4.2.2.1 Measurement procedure

The test shall be performed in three steps:

a) Measure the air volume flow through the air handling unit, external static pressure difference, the rotational speed of the air handling unit and effective electric input power of air handling unit drive analogically as in 5.2 for each air stream.

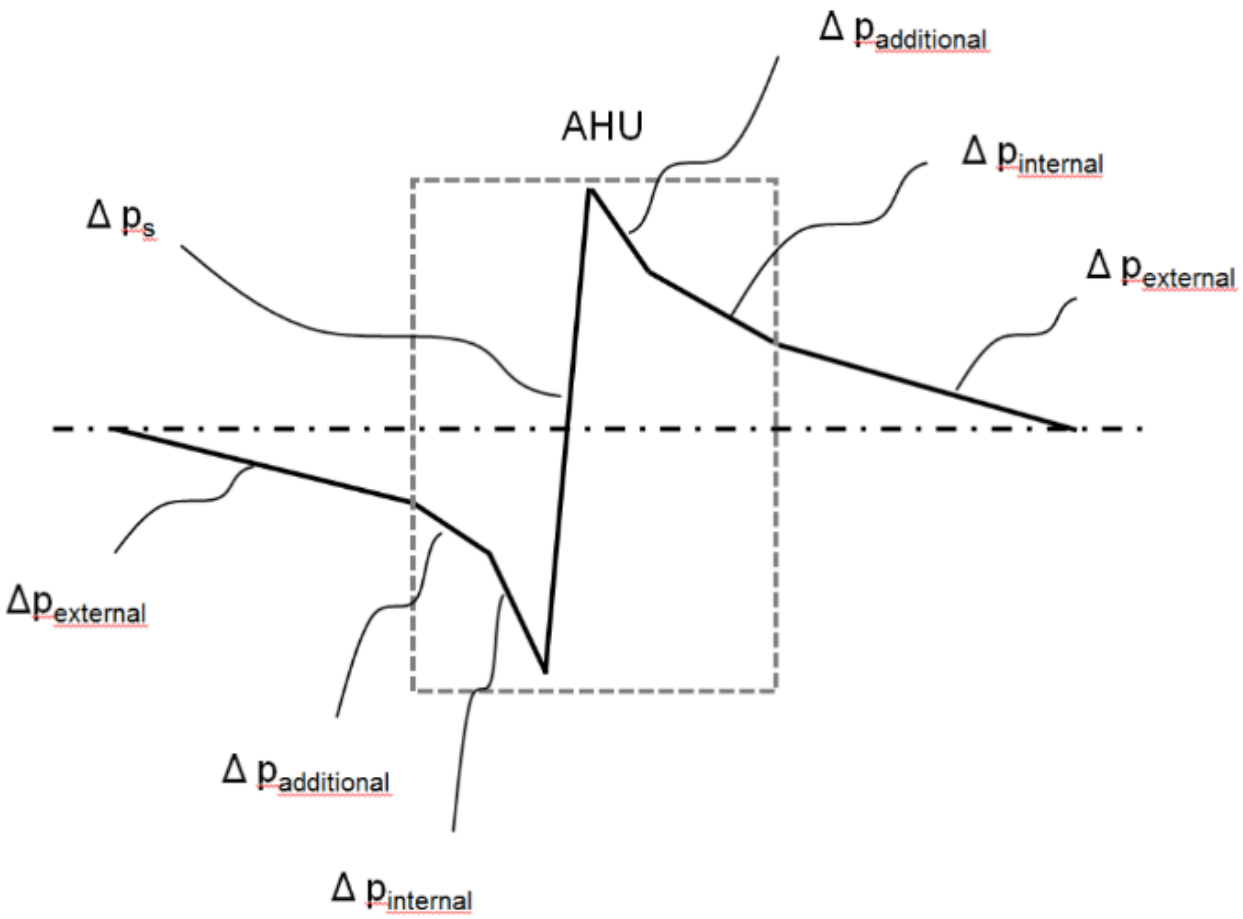

Figure 8 — Pressure diagram for the reference condition

Figures 8 and 9 show the static pressure measured at different places of an AHU.

b) Dismount all additional (ie. non-reference configuration) components and repeat the measurements in a), the fan speeds shall remain the same. Pressure losses of the dismounted additional components shall be added to the external pressure by adapting the external pressure by e.g. dampers on suction and / or pressure side (depending on the location of the additional components before or after the fan section). Measure air volume flow through the fan and external static pressure difference analogically as in 5.2 for each air stream.

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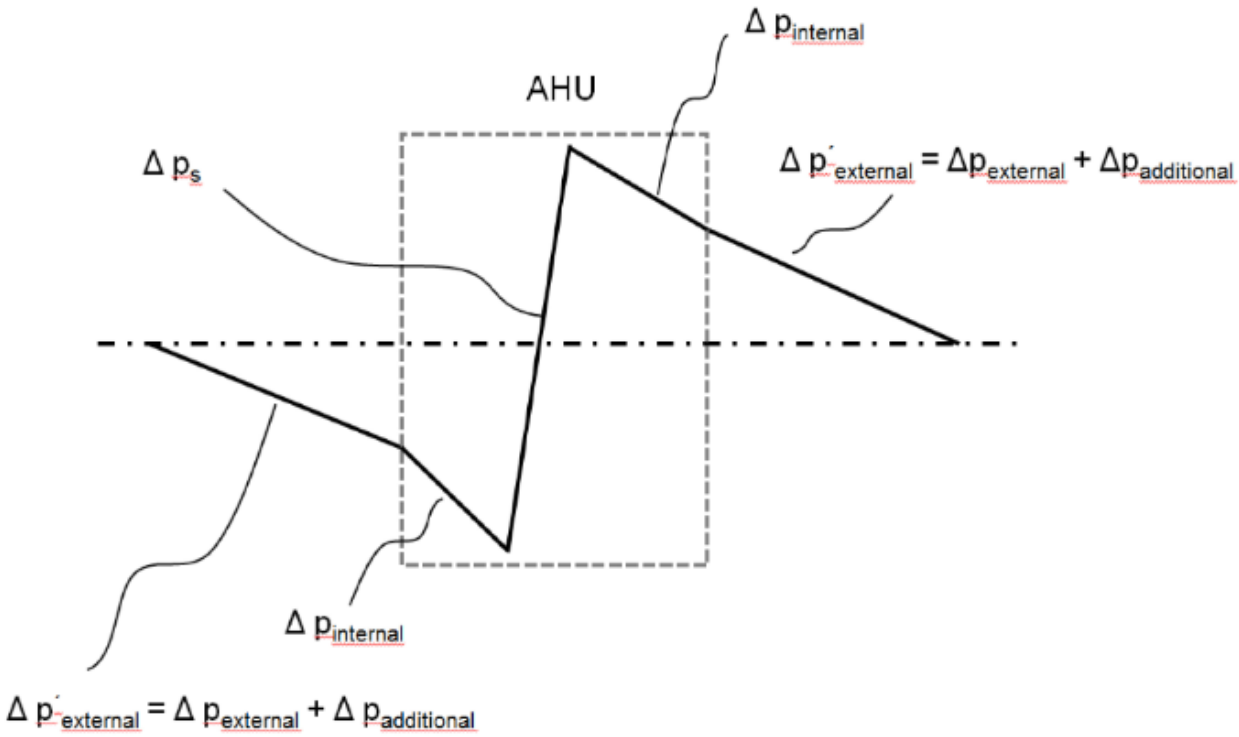

Figure 9 — Pressure diagram after dismounting of additional components and adjusted external pressure losses

c) The static pressure of the stand-alone fan (outside the unit) $\Delta p_{s,free,fan}$ shall be determined for fans in both air streams of the air handling unit by a test carried out in accordance with EN ISO 5801. The static pressure of the fan (stand-alone) corresponds to the fan static pressure (p_{fs}) according to EN ISO 5801 and shall be determined at the same air volume flow rate measured under a). The fan speed shall remain the same as in a) ± 5 %. If the difference between the fan speed inside and outside the unit is larger than ± 5 %, then the measurement values shall be corrected by laws of fan similarity according to EN ISO 5801.

Corrections in accordance with 5.2 shall be done.

5.4.2.2.2 Built-in fan efficiency

Built-in fan efficiency, $\eta_{s,fan}$ is overall static efficiency including motor and drive efficiency of the stand-alone fan.

$$\eta_{s,fan} = \frac{q_v \cdot \Delta p_{s,free,fan}}{P_{el}} \tag{14}$$

where

q_v	is the air flow rate expressed in $m^3 \cdot s^{-1}$
P_{el}	is the electical power consumption of fan (input power shall include speed control device if existing) expressed in (W)
$\Delta p_{s,free,fan}$	is the static fan pressure of the stand-alone fan, expressed in Pa.

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5.4.2.2.3 Evaluation of SFP_{int}

For air handling units with one air side:

$$SFP_{int} = \frac{\Delta p_{s,free,fan} - \Delta p_{s,ext}}{\eta_{s,free,fan}} \cdot \frac{P_{el}}{P_{el,free,fan}} \tag{15}$$

where

$$\eta_{s,free,fan} = \frac{q_v \cdot \Delta p_{s,free,fan}}{P_{el,free,fan}} \tag{16}$$

gives

$$SFP_{int} = \left(1 - \frac{\Delta p_{s,ext}}{\Delta p_{s,free,fan}}\right) \cdot \frac{P_{el}}{q_v} \tag{17}$$

where

SFP_{int}	is the internal specific fan power of ventilation components, expressed in $W \cdot m^{-3} \cdot s$;
$\Delta p_{s,ext}$	is the external static pressure difference of the unit from 5.4.2.2.1 b), expressed in Pa;
$\Delta p_{s,free,fan}$	is the static fan pressure of the stand-alone fan from 5.4.2.2.1 c), expressed in Pa;
P_{el}	is the effective electric input of the fan drive (input power shall include speed control device if existing) from 5.4.2.2.1 a), expressed in Watts [W];
$P_{el,free\ fan}$	is the effective electric input of the fan drive (input power shall include speed control device if existing) of the stand-alone fan, expressed in Watts [W];
$\eta_{s,free,fan}$	is the overall static efficiency including motor and drive efficiency of the stand-alone fan
q_v	is the air volume flow rate from 5.4.2.2.1 a), expressed in cubic meters per second ($m^3 \cdot s^{-1}$).

For air handling units with two air sides:

$$SFP_{int} = \left(1 - \frac{\Delta p_{s,ext,SUP}}{\Delta p_{s,free,fan,SUP}}\right) \cdot \frac{P_{el,SUP}}{q_{v,SUP}} + \left(1 - \frac{\Delta p_{s,ext,EHA}}{\Delta p_{s,free,fan,EHA}}\right) \cdot \frac{P_{el,EHA}}{q_{v,EHA}} \tag{18}$$

where

SFP_{int}	is the internal specific fan power of ventilation components, expressed in watts per cubic metre per second ($W \cdot m^{-3} \cdot s$);
Indexes	SUP is the supply air stream and EHA is the exhaust air stream.
$\Delta p_{s,ext,xxx}$	is the external static pressure difference of the unit in the air stream of interest (xxx is SUP or EHA) from 5.4.2.2.1 b), expressed in pascals (Pa);
$\Delta p_{s,free,fan,xxx}$	is the static fan pressure of the stand-alone fan in the air stream of interest (xxx is SUP or EHA) from 5.4.2.2.1 c), expressed in pascals (Pa);
$P_{el,xxx}$	is the effective electric input of the fan drive (including speed control device) of the air stream of interest (xxx is SUP or EHA) from 5.4.2.2.1 a), expressed in watts (W);

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$q_{v,xxx}$ is the air volume flow rate of the air stream of interest (xxx is SUP or EHA) from 5.4.2.2.1a), cubic meters per second ($m^3 \cdot s^{-1}$).

5.4.3 Measurement uncertainties

5.4.3.1 General

The following calculations are made under assumptions that all input values are independent.

5.4.3.2 Efficiency

The combined standard uncertainty for $\eta_{s,fan}$ results from the partial derivatives with:

$$u_{\eta_{s,fan}} = \sqrt{\left(\frac{\partial f\left(\eta_{s,fan}\right)}{\partial P_{el}}\right)^2 u_{P_{el}}^2 + \left(\frac{\partial f\left(\eta_{s,fan}\right)}{\partial q_v}\right)^2 u_{q_V}^2 + \left(\frac{\partial f\left(\eta_{s,fan}\right)}{\partial \Delta p_s}\right)^2 u_{\Delta p_{s,fan}}^2} \tag{19}$$

for

$$\eta_{s,fan} = \frac{q_v \cdot \Delta p_{s,fan}}{P_{el}} \tag{20}$$

for the absorbed motor power

$$\frac{\partial f\left(\eta_{s,fan}\right)}{\partial P_{el}} = -\frac{q_V \cdot \Delta p_{s,fan}}{P_{el}^2} \tag{21}$$

for the airflow

$$\frac{\partial f\left(\eta_{s,fan}\right)}{\partial q_v} = \frac{\Delta p_{s,fan}}{P_{el}} \tag{22}$$

for the static pressure difference

$$\frac{\partial f\left(\eta_{s,fan}\right)}{\partial \Delta p_{s,fan}} = \frac{q_v}{P_{el}} \tag{23}$$

5.4.3.3 Internal Specific Fan Power – SFP_{int}

The combined standard uncertainty for *SFP* results from the partial derivatives with:

$$u_{SFP_{int}} = \sqrt{\left(\frac{\partial f\left(SFP_{int}\right)}{\partial \Delta p_{s,int}}\right)^2 u_{\Delta p_{s,int}}^2 + \left(\frac{\partial f\left(SFP_{int}\right)}{\partial \eta_{s,fan}}\right)^2 u_{\eta_{s,fan}}^2} \tag{24}$$

for

$$SFP_{int} = \frac{\Delta p_{s,int}}{\eta_{s,fan}} \tag{25}$$

for the pressure difference

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$$\frac{\partial f\left(SFP_{\text{int}}\right)}{\partial \Delta p_{\text{s,int}}} = \frac{1}{\eta_{\text{s,fan}}} \tag{26}$$

for the efficiency

$$\frac{\partial f\left(SFP_{\text{int}}\right)}{\partial \eta_{\text{s,fan}}} = -\frac{\Delta p_{\text{s,int}}}{\eta_{\text{s,fan}}^2} \tag{27}$$

Alternatively:

$$u_{SFP_{\text{int}}} = \pm\sqrt{\left(\frac{\partial f\left(SFP_{\text{int}}\right)}{\partial P_{\text{el,int}}}\right)^2 u_{P_{\text{el,int}}}^2 + \left(\frac{\partial f\left(SFP_{\text{int}}\right)}{\partial q_{\text{V}}}\right)^2 u_{q_v}^2} \tag{28}$$

for

$$SFP_{\text{int}} = \frac{P_{\text{el,int}}}{q_{\text{v}}} \tag{29}$$

for the absorbed motor power

$$\frac{\partial f\left(SFP_{\text{int}}\right)}{\partial P_{\text{el,int}}} = \frac{1}{q_{\text{v}}} \tag{30}$$

for the airflow

$$\frac{\partial f\left(SFP_{\text{int}}\right)}{\partial q_{\text{v}}} = -\frac{P_{\text{el,int}}}{q_{\text{v}}^2} \tag{31}$$

NOTE EN ISO 5801 defines maximum allowable uncertainties for measurement of pressure, airflow rate and electrical power.

5.5 Tolerances

The air performance quoted or specified shall be the most probable, not the minimum or maximum acceptable value. The test for the specified duty shall be conducted in accordance with EN ISO 5801.

The tolerance should be applied to a specified duty or duties, not to every point on the air handling unit characteristic. The characteristic is drawn from the measured data and mathematically converted to the standard density 1,2 kg · m^{-3}. The tolerances to define the acceptability of an air handling unit are given In Table 2, in which the values correspond to tolerance grade AN 3 defined in ISO 13348.

The permissible deviation of the specified design point from the duty point on the air handling unit characteristic is the sum of the tolerance range of the specified duty point and the uncertainty range of the measured data. This uncertainty range derives from the measuring uncertainty of the methods of measurement and the measuring instruments and test rig used and shall be stated for a confidence level (probability) of 95 %.

For example, the departure for air volume flow is indicated in Figure 10.

where

t is the tolerance range of duty point, expressed in %;

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u	is the uncertainty range of measured data, expressed in %;
q_{vs}	is the specified air volume flow, expressed in $m^3 \cdot s^{-1}$;
q_{vm}	is the measured and converted air volume flow, expressed in $m^3 \cdot s^{-1}$;
$q_{vm} - q_{vs} = \Delta q_v \leq t \cdot q_{vs} + u \times q_{vm}$	is the allowable difference in air volume flow, expressed in cubic meters per second ($m^3 \cdot s^{-1}$).

Table 2 — Air handling unit performance tolerances

Working values	Tolerance range t	Remarks
Air volume flow q_v in $m^3 \cdot s^{-1}$	± 5 %	$\Delta q_V = (t_{qv} / 100\ \%) \cdot q_v$
Unit pressure p_u in Pa	± 5 %	$\Delta p_u = (t_{\Delta p}/100\ \%) \cdot p_u$
Unit static pressure p_{us} in Pa	± 5 %	$\Delta p_{us} = (t_{\Delta p}/100\ \%) \cdot p_{us}$
External static pressure difference $\Delta p_{s,ext}$	± 5 %	$\Delta(\Delta p_{s,ext}) = (t_{\Delta p}/100\ \%) \cdot \Delta p_{s,ext}$
Electrical motor input power P_{el} in W[a]	+ 8 %	$\Delta P_{el} = (t_P/100\ \%) \cdot P_{el}$ Negative deviations are permissible.
Total sound power level emitted to the ductwork and by the casing L_{WA} in dB	+ 4 dB	$\Delta L_{WA} = t_{LWA}$ The value t_{LWA} in dB is identical to the numerical value for the deviation limit for sound power level of the sound power level stated in dB(A). Negative deviations are permissible.

NOTE Uncertainties of the measured data, measuring instruments, and methods are considered in EN ISO 5801 and ISO 5168. An example is given in Figure 10.

[a] A simultaneous tolerance range of 5 % on air volume performance as well as external total pressure difference is acceptable. For electrical motor input power 8 % tolerance range at rated performance is allowed. Consequently, the measured electrical input power at a duty point deviating from the specified value should be converted to a value corresponding to the rated performance. A proportional relationship between input power and air volume flow and/or external total pressure difference may be assumed

5.6 Test report

The test report shall include the following information. The following list can be applied for testing any component or section with the relevant items completed and with additional items defined for the component or section:

a) date of the test;

b) name and location of the test laboratory;

c) names of the testing person;

d) type number and description of the air handling unit tested, including details from its rating plate, and configuration of the unit including filter and additional components;

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e) test standard applied;

f) test method and configuration adopted;

g) description and sketch of the air handling unit and test facility used including the position(s) of damper(s) in the unit;

h) detailed description of the joints between the unit and the ductwork;

i) identification of the instruments used;

j) tabulations of all measured quantities and the calculated values derived from them; the acoustic data shall be supplemented by the following information: Duty point of the unit including fan speed, air volume flow, unit pressure or unit static pressure, duct area(s), measurement standard(s) used, description of the test set-up. The acoustic data shall consist of the sound power levels, 100 Hz – 10 000 Hz in one-third octave band, or 125 Hz – 8 000 Hz in octave bands in dB, the overall value in dB(A) and end correction values if applicable;

k) tabulation of the correction for pressure difference between the measured clean filter pressure difference and that for the intermediate or final condition (if appropriate) and

l) graphs showing the unit pressure or unit static pressure and the fan electrical motor input power as functions of the air flow rate.

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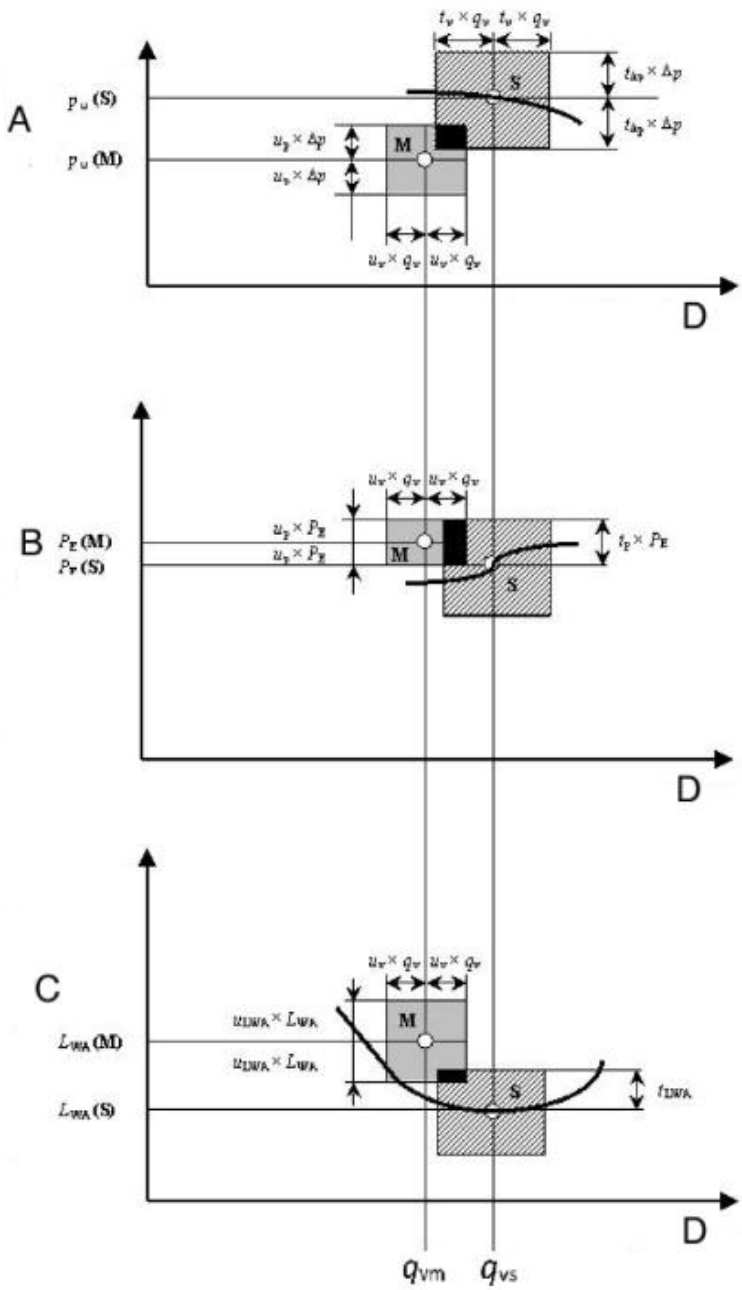

Key

A	unit pressure or unit static pressure	B	electrical motor input power
C	A-weighted total sound power level (L_{WA})	D	volume flow rate
M	measuring point	t	imit of deviation from agreed duty points
S	specified duty point	u	measuring uncertainty of measured variable

Figure 10 — Assessment of the data measured in a performance measurement against the agreed duty points

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6 Ratings and performance of components and sections

6.1 General

This chapter presents technical requirements and test methods, which shall be applied to components and sections of complete air handling units. However, it should be noted that the characteristics of a component or section when tested as a part of a complete air handling unit can be significantly different from those of the same component or section tested in ideal conditions as a stand-alone component.

Further guidance concerning the energy performance of the air handling unit and its components and sections is presented in EN 16798-3. This guidance includes examples of pressure differences for specific components in supply and extract air systems, in order to achieve a certain category for fan power consumption.

Annex A proposes examples of requirements and testing of defrosting arrangements of heat recovery sections.

Instructions for maintenance including recommendations for cleaning intervals, methods and equipment to be used shall be provided.

6.2 Casing

The equipment casings shall be made from corrosion-protected and abrasion-resistant materials, which neither emit substances which are harmful to health nor form a nutrient substrate for microorganisms. The wall structure shall consist of minimum of double skin panels with sandwiched insulation. The surface of the casing shall correspond at least to the quality level, e.g. galvanised steel sheet. Sharp edges or pointed objects shall be avoided.

The ingress of unfiltered air through casing leakage can cause hygiene problems. Therefore, the casing air tightness shall comply with the requirements specified in EN 1886.

It shall be possible to inspect, clean and disinfect all the components at a justifiable technical expenditure. Therefore, all equipment components shall be designed in a way that they are easily accessible and able to be cleaned from the operating side through upstream and downstream access doors or inspection panels or alternatively, they shall be able to be removed up to an interior height of 1,6 m. The seals used shall not absorb any moisture and shall not form a nutrient substrate for microorganisms. Cleaning the equipment requires smooth surfaces inside the casing.

Weatherproof equipment shall have inlet and outlet apertures with suitable weatherproof devices which provide protection from the weather even when the air handling unit is not running. In addition, outdoor air intake chambers shall be provided with a pan (quality of floor surface minimum galvanised and coated/painted steel sheet, powder coated, or wet painted with primer and top coat of thickness ≥ 60 µm or coil coated galvanised steel sheet) with a downward gradient for drainage to permit any entering water to drain away in a controlled way. Any equivalent equipment may also be used.

To avoid water entering the casing, the following maximum air velocities are recommended, see Table 3.

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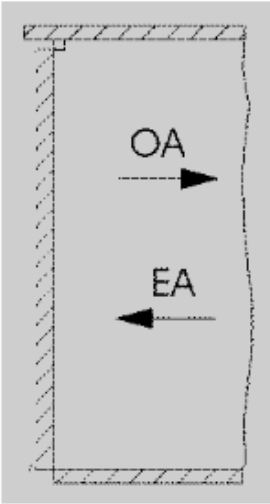
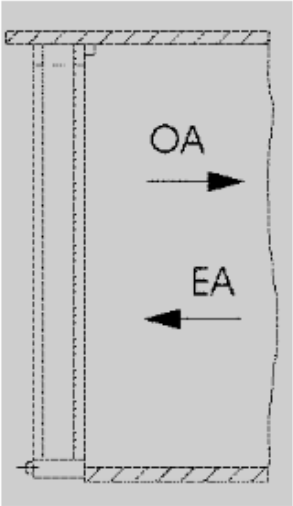
Table 3 — Weatherproofing / Recommended maximum air velocity

Weatherproofing		Recommended max. air velocity (with reference to the connection cross-section)	
		outdoor air side $m \cdot s^{-1}$	**exhaust air side** $m \cdot s^{-1}$
Louvre		2,5	4,0
Droplet eliminator		3,5	5,0

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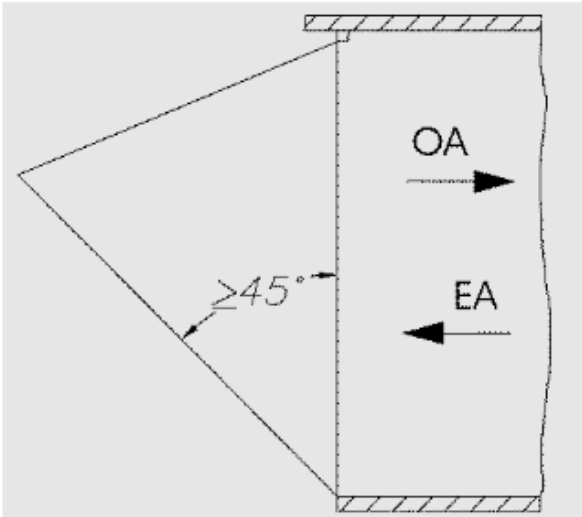
Weatherproofing		Recommended max. air velocity (with reference to the connection cross-section)	
		outdoor air side $m \cdot s^{-1}$	**exhaust air side** $m \cdot s^{-1}$
Rainhood		4,5	6,0

All apertures shall be protected by a grid to prevent the entry of small animals and coarse dirt (maximum of 20 mm × 20 mm).

Weatherproof air handling units shall not take over any static tasks or replace the function of the building.

NOTE 1 Very small grid holes are not recommended because of the effects of clogging.

NOTE 2 In cold climates it can be necessary to have a water-tight plenum section between the outdoor opening and the unit (or the first section), which guides the water immediately out of the building and/or is connected to a drain.

NOTE 3 Cold bridges in cabinets introduce a risk of condensation on the inner or outer surfaces, depending on which side of the unit is colder. Therefore, the thermal bridging factor class, as defined in EN 1886, will preferably be selected to take into account the climatic conditions in which the unit is expected to operate (e.g. particularly intake air chambers, cold coolers and weatherproof units).

NOTE 4 In cold climate it can be necessary to have an additional heating facility to prevent freezing of the inlet surface.

Table 4 — Classes of average air velocity levels inside the casing

Class	Air velocity $m \cdot s^{-1}$
Class V1	≤ 1,6
Class V2	> 1,6 to 1,8
Class V3	> 1,8 to 2,0
Class V4	> 2,0 to 2,2
Class V5	> 2,2 to 2,5

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Class	Air velocity $m \cdot s^{-1}$
Class V6	> 2,5 to 2,8
Class V7	> 2,8
NOTE The air velocity in the unit has a large influence on energy consumption. The velocities are calculated for air velocity in AHU cross-section. The velocity is based on the square area of filter section of a unit, or if no filter is installed, it is based on the square area of the fan section.	

NOTE 5 To reduce energy consumption it is strongly advised to reduce the velocity.

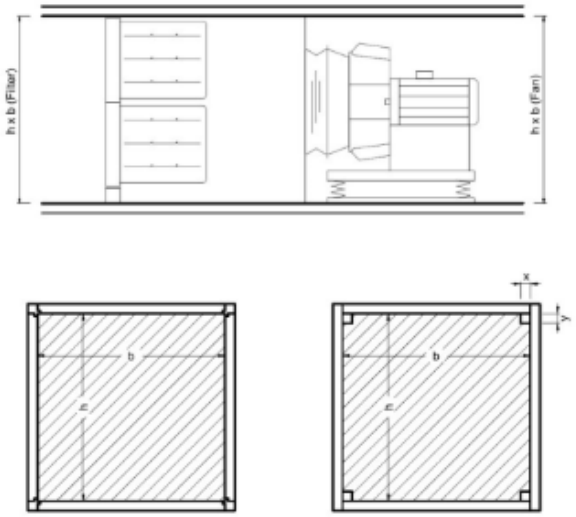

Figure 11 — Cross section of an Air Handling Unit

The Figure 11 shows a cross section (hatched area) of an Air Handling Unit with dimensions to be used in calculating the cross-section area.

6.3 Fan section

6.3.1 General

For hygiene reasons and to reduce maintenance expenditure, it is recommended to arrange the air supply fans so that suction-side leakage air flows are minimized. The arrangement of the fan in the air handling unit casing shall ensure an even inflow and outflow of air. Additional inflow and outflow devices should be fitted for this purpose if necessary.

The dynamic pressures on entry and exit should be low on economic aspects. The air velocity class in the unit shall be defined according to Table 4.

Fans with blades curved backward should be provided for energy reasons. To further reduce the consumption of electric power, energy-saving motors with increased efficiency should preferably be fitted.

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Starting from an interior height of 1,6 m, the fan chamber should be fitted with an inspection window (sight glass, inside diameter minimum 150 mm) and with a light.

A lockable maintenance switch shall be placed outside the air handling unit, near the fan section access door.

When selecting a fan for an air-handling unit, the pressure loss allowed for filters shall be in accordance with 6.8.2 and for the cooling coil pressure loss, the dry coil value shall be used unless otherwise stated.

6.3.2 Power input of fans

The power input of drives can be defined in classes. The maximum power input shall be calculated by the following formula:

$$Pm_{ref} = \left(\frac{\Delta p_{s,fan}}{450}\right)^{0,925} \cdot \left(q_v + 0,08\right)^{0,95} \tag{32}$$

where

Pm_{ref} is the reference power input, [W];

$\Delta p_{s,fan}$ is the static pressure difference over the fan to be measured at the fan section, [Pa];

q_v is the air flow of the fan, [$m^3 \cdot s^{-1}$].

Table 5 defines the power input ($P_m$max) classes:

Table 5 — Classes of power input of drives (fans)

Class	*Pm* max [kW]
Class P1	$\leq Pm_{ref} \cdot 0,85$
Class P2	$\leq Pm_{ref} \cdot 0,90$
Class P3	$\leq Pm_{ref} \cdot 0,95$
Class P4	$\leq Pm_{ref} \cdot 1,00$
Class P5	$\leq Pm_{ref} \cdot 1,06$
Class P6	$\leq Pm_{ref} \cdot 1,12$
Class P7	$> Pm_{ref} \cdot 1,12$
NOTE It is essential that each fan be specified in the power consumption classes. All values are based on nominal conditions with a density of 1,2 kg · m^{-3}.	

NOTE The power input *Pm* is the useful power supplied from the mains (including any motor control equipment).

6.4 Coils

6.4.1 General

This clause defines the requirements for coils used in air handling units. It applies to all coils, except electric heating coils.

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Coils are used for the thermodynamic treatment of air and in heat recovery systems. They shall be made from corrosion-resistant materials and the fins shall be easily cleanable, e.g. smooth fins.

6.4.2 Testing

Coils used in air-handling units shall be rated in accordance with EN 1216.

The capacity of the coil is determined by the difference in enthalpy and the flow rate on the water side.

As a confirming method, the capacity of the coil can be determined on the air side. As the coil under test is fitted into the air handling unit, air temperatures (dry bulb and wet bulb) shall be measured at the inlet and outlet of the air handling unit. Therefore, the capacity of the heating coil measured on the air side is given as the air enthalpy difference multiplied by the air mass flow rate from which the power input to the fan is subtracted.

In a similar manner, the capacity of the cooling coil is determined by the difference in air enthalpy multiplied by the air mass flow rate to which the power input to the fan is added. Measured outlet enthalpy shall be corrected for enthalpy in condensate flow in case of dehumidification.

The energy balance between the capacities measured on water and air sides shall, according to EN 1216, not be greater than 5 %.

6.4.3 Construction

For hygiene reasons cleaning, shall be guaranteed right through to the core. Therefore, heat exchangers shall be designed so that they are divided in the direction of the air if necessary (the maximum fin depth shall be 300 mm per heat exchanger stage, 450 mm for aligned tubes).

For energy and hygiene reasons the distance between the fins of the coolers that can dehumidify shall be at a minimum 2,5 mm, otherwise, the distance between fins shall be at a minimum 2,0 mm. In case of larger distances between the fins the mentioned depths can be expanded linearly.

Air heaters, which are used for drying before the first filter stage, shall guarantee a minimum distance between the fins of at least 4 mm.

To avoid significant bypass leakage, each heat exchanger shall be sealed within the casing of the air handling unit by means of sealing strips.

6.4.4 Cooler/Droplet eliminator

The same requirements for drainage, cleaning, materials and disinfection apply as for humidifiers.

For cooling coils that are designed to dehumidify, the following points shall be observed:

a) No moisture can carry over to the components or sections downstream of the coil.

b) For hygiene reasons, coolers with dehumidification shall not be arranged immediately before air filters or silencers. Fans or heaters shall be installed in between to limit the relative humidity.

c) Coolers shall be fitted with a corrosion-resistant drip pan (e.g. min. AISI 316 (stainless steel 1.4301) or corrosion-resistant aluminium alloy (minimum AlMg), which has a gradient towards the drain to permit unhindered drainage of condensate.

d) The connecting pipes shall be insulated where they pass through the casing, so that there will be no condensate from them.

e) For hygienic and energy reasons, droplet eliminators should only be used if the air velocity through the cooler does not exclude drops being carried over. They shall be designed in a way that they are easy to remove and dismantle without affecting any of the other unit components.

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f) It shall be possible to clean the cooling coil from both sides when in the mounted position or alternatively, if it is up to an internal height of 1,6 m, it shall be removable for cleaning purposes.

g) On corrosion-resistance aspects, a header of copper is recommended in case of copper/copper or copper/aluminium execution. If galvanised steel coolers are used, hot dip galvanising steel is recommended.

6.5 Heat recovery sections

6.5.1 General

Air handling units with supply and extract air and heating function should be fitted with heat recovery systems. When positioning heat recovery equipment, care shall be taken to minimize air leakage and unacceptable recirculating air flows.

The heat recovery system shall have a thermal by-pass facility.

To reduce the need for the use of mechanical cooling in the summer, in addition to heat recovery, evaporation cooling should be installed on the extract air side. The necessity for a condensate pan should be examined. If a condensate pan is necessary, the relevant requests (see 6.3.4) shall be complied with.

For reference a physical by-pass can reduce power consumption.

6.5.2 Classifications and requirements

The tests for thermal performances shall not be made when internal leakages of the HRS section, rated according to EN 308 are too high. HVAC-systems with supply and exhaust air should be fitted with heat recovery units. The minimum dry heat recovery efficiency (with reference to the mass flow ratio 1:1) and the maximum pressure losses are based on the annual running time of the system and the maximum outdoor air flow necessary during winter operation, see Table 5. The following characteristic values define the thermal efficiency of a heat recovery system (HRS) under balanced mass flows conditions (1:1). The following values should be indicated, η_e, η_t and class, see Table 6.

Further information is provided in Annex B.

Temperature efficiency (η_t) under dry condition:

$$\eta_t = \frac{t_{22} - t_{21}}{t_{11} - t_{21}} \tag{33}$$

where

t_{22} is the temperature of the supply air, [°C];

t_{21} is the temperature of the outdoor air, [°C];

t_{11} is the temperature of the extract air, [°C].

Pressure losses of the heat recovery system:

$$\Delta p_{HRS} = \Delta p_{SUP} + \Delta p_{EHA} \tag{34}$$

where

Δp_{HRS} is the sum of pressure losses, supply and exhaust, of the HRS, [Pa];

Δp_{SUP} is the pressure loss of the HRS supply air, [Pa];

Δp_{EHA} is the pressure loss of the HRS exhaust air, [Pa].

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NOTE 1 It is advised to consider all HRS based pressure changes (e.g. add. filters).

Electric power input (P_{el}) based on pressure losses:

$$P_{el} = q_v \cdot \Delta p_{HRS} \cdot \frac{1}{\eta_D} + P_{el,aux.} \tag{35}$$

where

q_v is the air flow, in $m^3 \cdot s^{-1}$ (standard density of 1,2 $kg \cdot m^{-3}$);

η_D is the average overall static efficiency of power consumption, [./.]; (if no specific value is declared, efficiency value 0,6 should be used)

$P_{el,aux.}$ is the auxiliary electric power input (e.g. pumps), [W].

NOTE 2 It is advised to consider all electric power consumptions influenced by the thermal capacity of HRS (e.g. water circulating pumps, air filters shall not be included).

NOTE 3 $P_{el,aux.}$ for pumps: $P_{el\ Pump} = q_v \cdot \Delta p_{HRS\ media} \cdot 1 / \eta_D$

Coefficient of performance (ε):

$$\varepsilon = \frac{\dot{Q}_{HRS}}{P_{el}} \tag{36}$$

Energy efficiency (η_e):

$$\eta_e = \eta_t \cdot \left(1 - \frac{1}{\varepsilon}\right) \tag{37}$$

The combined values (ε and η) shall be indicated at the following operating temperature conditions according to EN 308: t_{21} = + 5 °C and t_{11} = 25 °C. These values are not valid for all operating conditions. The primary energy influence is not considered, since it concerns a reference value contrary to annual values.

Table 6 defines the heat recovery classes at balanced mass flow conditions (1:1):

Table 6 — Classes of heat recovery

Class	$\eta_{e\ 1:1}$ min [%]
Class H1	≥ 74
Class H2	≥ 70
Class H3	≥ 65
Class H4	≥ 60
Class H5	< 60
NOTE The values are valid for balanced mass flows (1:1). The classes define the quality of the HRS and they have a strong influence on the thermal energy consumption. In Nordic countries, higher classes and in southern countries lower classes are common.	

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Yearly energy computations, depending on the location and the mode of operation of the heat recovery unit should supply the reference values for the necessary classification based on Table 6. National regulations for the determination of the heat recovery efficiency may deviate from this classification.

NOTE 4 If the airflows are not balanced and no specific HRS values are available, the values may be calculated by the empiric formula (η_t 1:1 is the value for balanced mass flow):

$$\eta_{t1:1} = \eta_t \cdot \frac{\left(1 + \frac{m_2}{m_1}\right)}{2}$$

This formula shall not be used to recalculate combined values. η_e should be recalculated with η_t and ε.

NOTE 5 The table is based on following calculation:

Class	η_t	Δp_{HRS} [Pa]	ε	η_e
H1	0,78	2 × 300	18,8	0,74
H2	0,73	2 × 240	22,0	0,70
H3	0,68	2 × 190	25,9	0,65
H4	0,63	2 × 150	30,4	0,60

Requirements for heat exchangers according to this standard are as follows:

a) all heat recovery sections should have 4 pressure tapping points, one on each air flow side of the exchanger;

b) within heat recovery sections fitted with category I and II heat exchangers there shall be a drain pan for condensate;

c) category III heat exchangers shall include a purge sector, except when recirculation air is used.

6.5.3 Testing

The heat recovery device and section shall be rated in accordance with EN 308.

The test shall be carried out without activating the thermal bypass.

Where relevant, the ability to function at low outdoor temperatures and the effectiveness of defrosting arrangements should be tested in accordance with Annex A.

The risk of frosting and the need for testing the defrosting arrangement shall be established by calculations which take into account air flows, including the balance between supply and extract air flow, type and temperature ratio of the heat exchanger, outdoor air temperature and extract air temperature and humidity. Internal leakages of the whole heat recovery section shall be measured according to EN 308.

6.5.4 Construction

For run around coil systems the requirements for coils in 6.4.3 apply. For hygienic reasons plate heat exchangers with a fin space of 3 mm should be installed in a manner that enables cleaning (e.g. installation in segments) when the depth exceeds 1 200 mm. In case of larger distances between the fins the maximum depth can be expanded linearly.

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6.6 Damper sections

6.6.1 General

Air regulating and shut-off dampers shall be tested according to EN 1751. The characteristics of dampers shall be classified in accordance with EN 1751. The face velocity shall be limited to 8 m · s^{-1} (exception: recirculation air and bypass dampers). On energy and function aspects, an inflow angle of minimum α = 25° and an outflow angle of minimum β = 35° is recommended (see Figure 11).

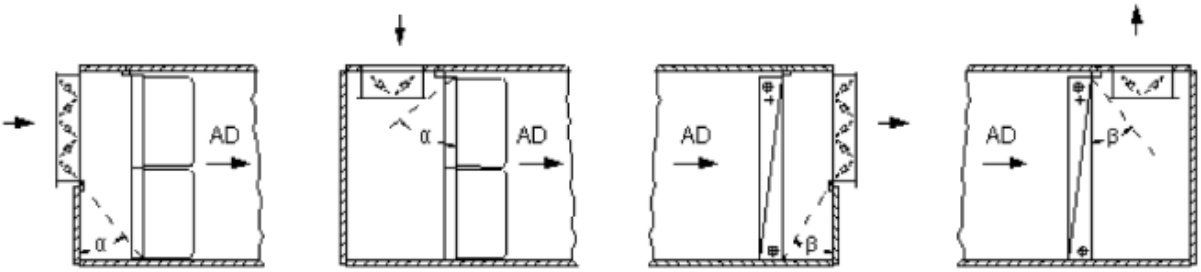

Figure 12 — Inflow and outflow angle

6.6.2 Requirements and testing

For all dampers which are intended to be closed completely during operation, e.g. bypass dampers for heat recovery sections and recirculation dampers for mixing section, the air tightness for the damper in a closed position shall meet the air tightness requirement of class 2 in accordance with EN 1751. For installations with high requirements for hygiene or energy economy, supply air and exhaust air-dampers shall meet the air tightness requirement of class 3.

6.7 Mixing sections

6.7.1 General

This clause specifies the requirements and testing methods for standard mixing sections used in air-handling units to mix and control air flows with different temperatures, primarily exhaust air with outdoor air, in order to recirculate it in a building.

6.7.2 Categories and characteristics

6.7.2.1 Category A - On-off sections, operating only in certain conditions (e.g. night heating)

Where the function utilizes 100 % outdoor-exhaust air or 100 % recirculated air, the characteristics to be specified are as follows:

— tightness of recirculation air damper;

— uniformity of flow after the mixing section or minimum distance to specified sections;

— pressure difference of dampers for calculating the difference between volume flows in "recirculation" and "closed" positions.

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6.7.2.2 Category B - Sections for flow control

Where the function utilizes control/mixing air flows, the characteristics to be specified are as follows:

a) damper tightness, uniformity of flow after the mixing section or minimum distance to downstream sections sensitive to non-uniform flow;

b) mixing characteristics according to 6.7.3;

c) temperature gradient (stratification);

d) risk of freezing;

e) risk of condensation and

f) pressure difference of dampers for calculation of difference of volume flows in different positions.

6.7.3 Requirements

6.7.3.1 General

The mixing section can have a major influence on the air flows and pressure balance within the ventilation or air conditioning system and hence the building. The quality of mixing is characterized by the temperature mixing efficiency specified in 6.7.3.2.

The mixing efficiency shall be measured at recirculation flow damper positions 90 % open, 50 % open, and 20 % open.

For assessing the lowest or highest possible air temperature imtmediately downstream of the mixing section, the mean temperature of the mixed flow can be derived from the quantities of outdoor and recirculated air flows in accordance with Formula (38).

The cross-sectional areas, temperatures, velocities and densities are as shown in Figure 12. Subscript "H" refers to the higher temperature air flow, "L" refers to the lower temperature air flow and "tot" to the air flow downstream of the mixing section.

$$t_M = \frac{t_{HH} \cdot \rho_{HH} \cdot q_{vHH} \cdot t_{LH} \cdot \rho_{LH} \cdot q_{vLH}}{\rho_{totH} \cdot q_{vtotH}} \tag{38}$$

The mean velocity is calculated using formula

$$V_M = \frac{q_v}{A_{tot}} \tag{39}$$

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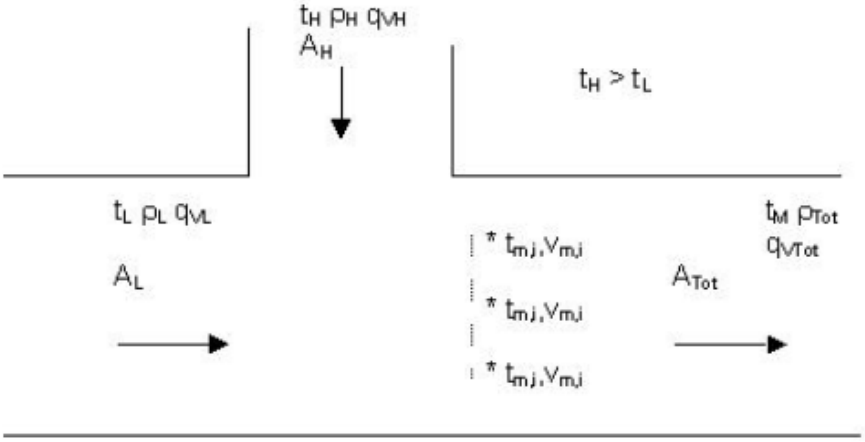

Figure 13 — Quantities to calculate the mean velocity and mean temperature and to define mixing efficiency

6.7.3.2 Temperature mixing efficiency

Measure the temperatures and velocities downstream of the mixing section according to 6.7.4 (see Figure 13).

The temperature mixing efficiency is calculated from formula

$$\eta_{mix} = \left(1 - \frac{t_{max} - t_{min}}{t_H - t_L}\right) \cdot 100\% \tag{40}$$

where

η_{mix} is the mixing efficiency, expressed in %;

t_{max} is the highest temperature in the measuring plane downstream the mixing section, expressed in °C;

t_{min} is the lowest temperature in the measuring plane downstream the mixing section, expressed in °C;

t_H is the higher temperature of entering air, expressed in °C;

t_L is the lower temperature of entering air, expressed in °C.

The classification of mixing temperature efficiency is given in Table 7.

Table 7 — Mixing temperature efficiency

Class	Mixing efficiency %
M1	≥ 95
M2	85 ≤ η < 95
M3	70 ≤ η < 85
M4	50 ≤ η < 70
M5	< 50

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6.7.3.3 Uniformity of flow after the mixing section

The uniformity of flow after the mixing section is calculated using Formula (41).

NOTE Because of the often very uneven and turbulent air flow immediately downstream of the mixing section, velocity measurements at these points may be very inaccurate and the results should be used only as a rough estimate of the velocity profile.

$$\frac{v_{\min}}{v_{\mathrm{m}}} \leq U \leq \frac{v_{\max}}{v_{\mathrm{m}}} \tag{41}$$

where

U	is the range of uniformity of flow; bounded by the lower and upper ratio of minimum and the maximum velocity to mean velocity;
$v_{\min}$	is the lowest velocity on a grid at the end of the mixing section, expressed in $\mathrm{m \cdot s^{-1}}$;
$v_{\max}$	is the highest velocity on a grid at the end of the mixing section, expressed in $\mathrm{m \cdot s^{-1}}$;
v_{m}	is the calculated mean velocity in the cross section at the end of the mixing section, expressed in $\mathrm{m \cdot s^{-1}}$.

6.7.4 Measurements

6.7.4.1 General

This method is applicable for rating purposes of stand alone mixing sections under laboratory conditions. For field tests, which are not applicable for rating purposes, see 6.7.5. The efficiency in a real air handling unit also depends on the configuration of the whole unit and how it is connected to the system.

6.7.4.2 Measurement of air temperature

Air temperatures shall be measured on a grid at the end of the mixing section just upstream of the position where a defined mixing quality shall be available. At least 3 × 3 temperature measuring devices should be installed equispaced in the vertical and horizontal directions of the grid. The distance between the casing and the nearest temperature measuring devices should exceed 25 mm and the distance between the adjacent temperature measuring devices should not be less than 100 mm but not more than 300 mm.

The distance between the casing and the nearest measuring points should be half the distance between the next points. For rating purposes, the temperatures of the two incoming air flows to be mixed shall differ by more than 25 K.

6.7.4.3 Measurement of air velocity

The local air velocities shall be measured at the same points as the temperatures, using probes with directional sensitivity to measure only axial velocity components in the grid, e.g. pitot static tube measurements in combination with high sensitive micromanometers.

6.7.4.4 Measurement of air flows

The air flow measurement shall be made on each side of the mixing section. Methods in accordance with EN ISO 5167-1, ISO 3966 or EN ISO 5801 shall be applied.

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6.7.5 Field testing of mixing efficiency

Field tests are often needed to check the functioning of the mixing section in real installations, e.g. during commissioning, during periodical inspections of the system or when an existing air handling unit is modified, by adding new sections or components. A field test is not valid for rating purposes.

The accuracy of the field testing of temperature mixing efficiency is decreased when the temperature difference between the two air flows is lowered. The field test shall be undertaken at as high a temperature difference as practically possible. When the temperature difference is less than 10 K, only the grade of uniformity shall be reported.

6.7.6 Condition for pressure loss measurement of ventilation component due to mixing

The design outdoor air flow rate when using a mixing section can be less than the design supply air flow rate which includes recirculation air flow. The fan shall be designed for the full air flow (e.g. including recirculated air flow for supply air fan). Pressure losses of ventilation components (heat recovery sections, filters, etc.) upstream of the mixing section in the supply air stream and downstream in the exhaust air stream shall be measured at their respective desig flow rates.

6.8 Humidifiers

6.8.1 General

Air humidification sections shall be operable in such a way that they do not cause any health hazards. The selection of materials to be used in the humidifier shall be made taking account corrosion-resistance, hygiene, bacteriostatic or bactericidal surface effect, ability of microbes to metabolize the material, resistance to disinfectants, ability to be cleaned and, if applicable, resistance to the respective disinfection process. The plastics used shall not contain sources of nutrition for microbiological growth.

For air handling units with supply air humidifiers (exception: steam humidifiers) it is recommended to be equipped with at least two filter stages, whereby the humidifier is to be arranged between the first and second filter stage. Humidifiers (exception: steam humidifiers) shall not be immediately arranged before air filters or attenuators. The first filter stage shall be ISO ePM1 50 % on hygiene aspects. The sealing materials used shall be made of a closed cell type and shall not absorb any moisture.

Only humidifier water containing bacteria in a concentration that is not detrimental to health is used for air handling purposes. If the humidifier water is suspected to contain more bacteria, it shall be checked for pathogenic bacteria.

The upper limit value for non-pathogenic bacteria is 10 000 cfu[1)] · ml^{-1}, however, from a bacteria content of 1 000 cfu · ml^{-1} onwards in the humidifier water, the plant should be checked and cleaned.

NOTE These are default values – national regulations, standards and guidelines may specify other limit values.

The manufacturer's maintenance instructions shall be available and observed.

In the case of humidifiers operating with recirculating water, from the point of view of reducing the number of bacteria, the dissolved solids content, and the dirt particles, it is better to empty the tray completely rather than to bleed off continuously.

Disinfectants can be used during cleaning after all the accumulated dirt has been removed.

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1) cfu means colony forming unit

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Sufficient overflow shall be arranged in evaporative humidifiers during operation. For nonoperation time, a complete emptying of the tray shall be done. Ultraviolet treatment and regular flushing are recommended.

6.8.2 Categories

Humidifiers are categorized according to type of construction as follows:

— spray humidifiers: A: air washers

B: ultrasonic humidifiers

C: high-pressure atomisers

— evaporative humidifiers: D: contact humidifiers

— steam humidifiers: E

6.8.3 Requirements

6.8.3.1 Droplet impingement on downstream components

In order to avoid droplets impinging on components downstream of the humidifier, the length of the humidifier section shall be sized accordingly, and/or suitable components for the separation of water (e.g. droplet eliminators) shall be installed.

6.8.3.2 Surface finish of the humidifier casing

Examples of surface materials for humidifier casing are as follows:

Categories A, C: inside stainless steel or corrosion-proof aluminium (minimum AlMg) or resin coated glass fibre.

Categories B, D, E: inside steel sheet, galvanised and coated (powder coating or two-coat wet painting with priming and finishing coats, minimum 60 µm) or galvanised steel sheet and stripcoated.

6.8.3.3 Constructional details

The requirements for construction for different categories are specified in Table 8.

Table 8 — Constructional details for different humidifiers

Item	Requirements	Humidifier category
1	Humidifier parts easily accessible through inspection door or panel for cleaning and maintenance purposes	A to D
2	Built-in parts such as droplet eliminators, nozzles and pipes dismountable	A to E
3	All water-carrying parts corrosion proof	A to E
4	Tray made of corrosion proof materials, e.g. stainless steel or aluminium (minimum AlMg)	A to E
5	Tray with all sides sloped, completely drainable	A, C, D
6	Inspection window (minimum diameter 150 mm) and internal light	B to E

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Item	Requirements	Humidifier category
7	Inspection window (min. diameter 150 mm) with blind, and internal light (minimum IP 65). Completely emptied and dried automatically (dry running device). Where lighting is fitted externally, care shall also be taken that when the light is switched off, no light can penetrate into the humidification chamber.	A
8	Dry-running protection device for pump	A, C, D
9	If the use of a disinfection processes is necessary to avoid the growth of germs, only methods should be used, the effectiveness of which has been proved in practice and which have been proved to be safe as regards health. Disinfectants shall not get into the room air through the humidification process.	A to E
10	Automatic bleed / sludge removal device	A, D
11	Interior casing watertight at negative pressure and positive pressure	A, C

6.8.3.4 Testing of adiabatic humidification

For efficiency testing of adiabatic humidification systems, the following physical dimensions shall be measured:

— air volume flow rate (q_v);

— mass flow of water inlet (q_w);

— mass flow of water drain and overflow (q_d);

— mean air temperature before the entrance of the humidifier (t_1);

— mean air temperature after the humidifier (t_2);

— wet-bulb temperature (saturation) (t_3);

The efficiency of humidification η_h is determined according to Figure 14.

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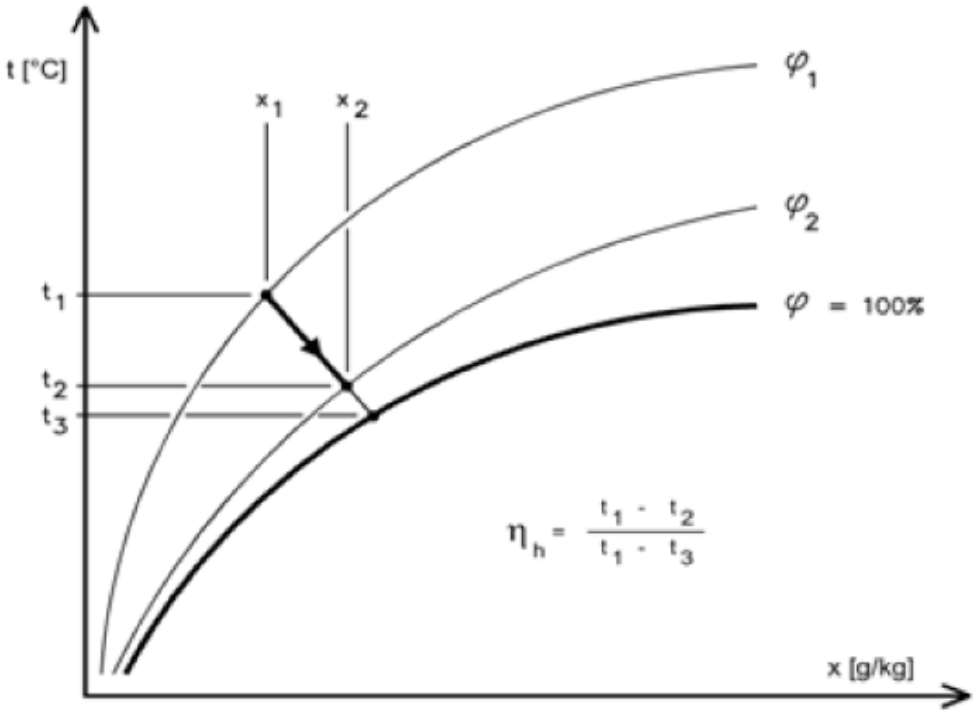

Figure 14 — Humidifier efficiency

In order to achieve a sufficient measuring accuracy, the temperature difference between the air temperature upstream the humidifier t_1 and the wet-bulb temperature t_3 should be at least 10 K. If necessary, air is to be heated before entering the humidifier.

Under the test conditions the balance between humidifier evaporation q_{hv} measured on the air side and the water side q_{hw} shall not be greater than 10 %.

$$q_{hv} = q_v \left(x_2 - x_1 \right) \tag{42}$$

$$q_{hw} = q_w - q_d \tag{43}$$

where

x	is the absolute humidity expressed in grams per kilogram ($g \cdot kg^{-1}$)
q_v	is the air volume flow rate, expressed in $m^3 \cdot s^{-1}$;
q_w	is the mass flow of water inlet, expressed in $kg \cdot s^{-1}$;
q_d	is the mass flow of water drain and overflow, expressed in $kg \cdot s^{-1}$.

The waste water factor is the ratio between the water flow not used for humidification and the supplied water to the humidifier. Efficiency should not be mixed with water consumption or waste water factor.

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6.9 Filter sections

6.9.1 General requirements

The task of air filters in HVAC systems is not only to protect the ventilated rooms from too severe a level of contamination but also the HVAC system itself. This is guaranteed by the use of fine filters of filter class group ISO ePM1, ISO ePM2.5 or ISO ePM10 according to EN ISO 16890. When manufacturing air filters, no components or materials may be used which can serve microbes as nutrients.

The requirements for air tightness, strength, and bypass leakage are specified in EN 1886.

The side wall on the service side of the filter section shall be equipped with an inspection door. The width and height of the door shall be greater than the external dimensions of the replaceable filter elements. There shall be a free space to the side of the access door and immediately upstream of front access filters sufficient to allow unrestricted access for filter removal and replacement.

The filter section shall be equipped with tappings for a pressure loss gauge/ manometer.

Additional requirements can be specified which take into account the climatic conditions (e.g. low temperatures, moisture, sand, and salt mist).

Additional measures are required in cold climates where the possible accumulation of frost will require the slight preheating of supply air and where there is excessive mist in outdoor air, the moisture running off the filters can necessitate specific requirements for corrosion protection.

6.9.2 Filters installed in air handling units

The first filter stage is to be fitted on the intake side, as close as possible to the outer air intake aperture to keep the air treatment elements as clean as possible. Additional coarse filters are permissible. The second filter stage is arranged on the output side at the beginning of the supply duct in order to keep the ductwork clean.

If a single stage filter system is used for supply air system, a filter of minimum ISO ePM1 50 % shall be fitted.

If two-stage filtering is used, the supply air fan shall be arranged between the first and second filter stage. To avoid microbial growth on air filters of the second or higher filter stage, the relative humidity in the area of the filter is to be limited to 90 %; dropping below the dew point in the area of the air filter shall always be avoided. Air filters shall not be arranged immediately after coolers with dehumidification or after humidifiers (exception – steam humidifiers).

If bag filters are used, the filter area should be at least 10 m^2 per 1 m^2 equipment cross-section. The seals used shall be of a closed cell type, shall not absorb any moisture and shall not form a nutrient substrate for microorganisms. A permanent tight fit shall be guaranteed for the seal (e.g. operation from the dusty air side). Starting from an interior height of 1,6 m, the filter chamber should be fitted with an inspection window (sight glass, inside diameter minimum 150 mm) and with light.

For fan selection purposes the filter pressure loss value at design volume flow shall be the average of the initial and final pressure losses for clean and dust loaded filters.

The filter section shall be equipped with measuring devices (e.g. signalling or alarm) for pressure difference.

NOTE 1 It is advised to give variation in volume flow caused by the accumulation of dust in technical specifications. If specific tolerances for an application are not specified, ± 10 % based on the average pressure difference is acceptable.

The pressure loss of a filter section loaded with dust shall not exceed the values given in Table 9. Lower final pressure differences can be also specified where appropriate.

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Filters installed in air handling units used for human occupancy shall be tested and classified according to EN ISO 16890.

Table 9 — Maximum final pressure difference for filters

Filter class	Final pressure difference
ISO coarse	The smaller value of either adding 50 Pa to the clean filter pressure difference or three times the pressure difference of clean filters.
ISO ePM1, ISO ePM2,5, ISO ePM10	The smaller value of either adding 100 Pa to the clean filter pressure difference or three times the pressure difference of clean filters.

NOTE 2 The final pressure differences tabulated in Table 9 are the typical maximum values for air-handling units in operation and lower than those used in EN ISO 16890 for classification purposes, for reasons of energy saving, and the performance obtained from tests according to EN ISO 16890 are not necessarily met at these lower pressure differences.

The following data shall be displayed in a clear, visible form (e.g. label) on the filter section: filter class, type of filter medium, final pressure difference. On changing the filter, the user shall check and update this information.

6.10 Passive sound attenuation sections

The performance of sound attenuation sections shall be tested according to EN ISO 7235.

Attenuators should be placed immediately adjacent to the source generating the noise to reduce noise emissions and should preferably be installed in the air handling unit immediately before and after the fan. On hygiene aspects, they shall not be arranged immediately after coolers with dehumidification or other humidifying devices.

To ensure unhindered inflow and outflow, a minimum distance from other installed components of 1,0 × (inflow) and/or 1,5 × (outflow) maximum splitter thickness shall be provided. The individual splitters should be removable for cleaning and shall consist of permanently abrasion-resistant material which is safe from a health point of view. No fibres shall be loosened during service. The use of splitter cones can reduce the pressure difference.

7 Extended hygiene requirements for special applications

7.1 General

Air handling units with high hygiene requirements (e.g. hospitals, clean rooms, pharmaceutical industries) shall also meet the requirements defined in this Clause 7.

7.2 Accessibility

The components of air handling units shall be accessible for cleaning purposes through access doors both upstream and downstream, or alternatively they shall be easy and safe to remove.

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7.3 Smoothness

Any half-closed profiles or joints that can accumulate pollutants and dirt and are difficult to clean, shall not be accepted, especially in the cabinet floor. All fibrous and porous material, except replaceable components like filter cells, shall be protected by suitable smooth material, which can withstand frequent cleaning. Screws and other similar components shall not protrude from the internal walls.

7.4 Inspection windows and lights

All units shall be provided with inspection windows and internal lighting for checking at least the fans, filters, and humidifiers.

7.5 Drainage/prevention of condensation, humidifiers

For non-pathogenic bacteria contained in the humidifier water used for air handling purposes, the upper limit value is 1 000 cfu × ml^{-1}. However, from a bacteria concentration of 100 cfu × ml^{-1} onwards in the humidifier water, the plant should be checked and cleaned.

NOTE These are default values – national regulations, standards and guidelines may specify other limit values.

It is reasonable to use ultraviolet sterilizers to reduce the number of bacteria. When designing and adjusting them, however, care shall be taken that no ozone is generated and enters the served space.

Biocides can only be used if, under no circumstances, they are detrimental to the health of the occupants in the areas served by the air handling unit.

7.6 Air leakage

For installations with high requirements for hygiene or energy economy, shut-off dampers for supply and exhaust air shall meet the air tightness requirement of class 4. The casing air leakage shall not exceed class L2 (R) according to EN 1886.

8 Instructions for installation, operation and maintenance

8.1 Installation

The air handling unit should be installed according to the manufacturer's instruction.

The instructions for installation and commissioning should be in accordance with existing standards, codes, and rules. The instructions should include information about the space required for maintenance, mounting and supports etc., preferably including detailed drawings and/or technical data. Connections to water, drainage, and the electrical supply network should also be presented in detailed drawings. The unit should be easy to connect to these networks and also easy to remove if service or repair is needed.

Air handling units shall be equipped with suitable lifting devices such as crane eyes, wood or pallets for transportation by crane or forklift.

Components at risk, e.g. fans on spring isolators, shall be protected by safety devices during transportation. A label should be attached to the unit stating that such devices shall be removed upon installation.

Particularly sensitive components or attachments in the area of division between pieces of equipment shall be protected from damage by suitable measures (e.g. freely accessible fin packages of heat exchangers should be completely covered).

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8.2 Operation and maintenance

Instructions for operation and maintenance should include:

— instructions for safe use in service and maintenance;

— instructions for starting and closing down the equipment;

— instructions for monitoring equipment and instrumentation, periodical inspections, recommendations for inspection intervals;

— description of normal operation of the unit, instructions concerning protection and control equipment, instructions for fault finding;

— service and cleaning instructions including drawings; for components which require periodical servicing or changing, an estimated service schedule and a list of spare parts and accessories are required;

— estimated schedule for periodical inspections.

For each functional section of the air-handling unit, appropriate instructions for operation and maintenance are required

8.3 Documentation and labelling

Air handling units shall have permanently attached type plates with permanent labelling.

In addition to the manufacturer, type and order number, all the necessary technical data shall also be clearly shown. A drawing of the unit with all the main and duct connection dimensions, a design data sheet, a spare part list plus assembly, commissioning and maintenance instructions shall be supplied with the air handling unit.

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Annex A
(informative)

Air handling units - Heat recovery – Defrosting - Requirements and testing

A.1 General

This annex covers the laboratory testing of the correct functioning and energy recovery of air handling units with air-to-air heat exchangers of category I or II according to EN 308 under conditions where frosting can occur. The tests are performed at specified duty points and the result can be used for comparisons and calculations of recovered heat during a longer period (normally one year).

NOTE For testing air-to-air heat exchangers, EN 308 describes a method for laboratory testing of leakage, pressure difference and temperature ratio. However, in cold climate heat exchangers of category I and II often can have frosting problems. Due to the fact that defrosting is a matter of, not only the heat exchanger itself, but also the whole air handling unit, this annex specifies a method for testing the defrosting and frost protection arrangements for air-handling units. Frosting can occur at low outdoor temperatures when moisture is added to the air in the building. The loss of recovered energy can be considerable. The type of heat exchanger, efficiency, and exhaust air temperature can also influence the amount of frosting problems. For cross-flow heat exchangers these problems typically occur at outdoor temperatures lower than −5 °C when moisture is added to the air, not only by the emission from human beings but also due to activities and processes such as, cooking, washing and drying.

A.2 Defrosting

A.2.1 Defrosting heat factor

$$\varepsilon_D = \frac{\sum_{i=1}^{k}\left[q_{m22,i} \cdot c_{p2} \cdot \left(t_{22,i} - t_{21,i}\right)\Delta\tau_i\right] - \dot{Q}_{defr}}{q_{m11}\sum_{i=1}^{k} c_{p1} \cdot \left(t_{11,i} - t_{21,i}\right)\Delta\tau_i} \quad (A.1)$$

where

ε_D is the defrosting heat factor;

n is the number of measurements within the total measuring time;

$\Delta\tau$ is the sampling interval time, expressed in s;

$\dot{Q}_{defr}$ is the total energy input for defrosting during one complete frosting/defrosting cycle, expressed in J.

A.2.2 Non-cyclic defrosting

The unit is provided with a continuously working defrosting function which stabilizes or avoids frost formation. The static pressure difference on the exhaust air side remains unchanged.

A.2.3 Cyclic defrosting

The unit allows for frost formation followed by a defrosting period. This results in a cyclic increase/decrease of pressure difference on the exhaust air side.

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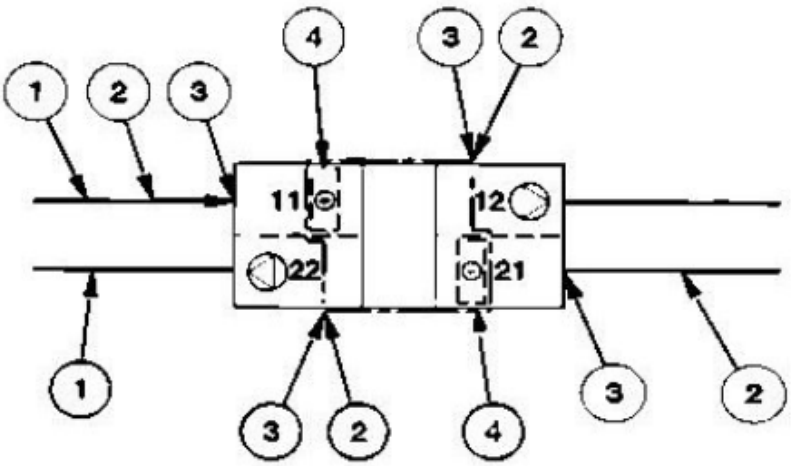
A.3 Testing

A.3.1 Test rig

The complete air-handling unit should operate with its own fans and be installed with external ducts to ensure that the defrosting equipment acts in a similar way to a real installation. The total external pressure difference should be 250 Pa at nominal air flow on both supply air side and exhaust air side. The pressure loss coefficient of the external parts of the test rig should be constant during the test, so that only frosting can influence the air flow. Ambient temperature should be (20 ± 3) °C, (see Figure A.1).



Key

1	Measurement of air flow	11	Extract air
2	Measurement of static pressure	12	Exhaust air
3	Measurement of temperature	21	Outdoor air
4	Defrosting energy	22	Supply air

Figure A.1 — Test arrangement for defrosting test

A.3.2 Duty points

Heat recovery performance should be determined for the following two duty points:

Air flow Supply air $q_{m2} = q_{mn}$

Exhaust air $q_{m1} = q_{mn}$

NOTE 1 These are the initial air flows. Both air flows can vary during the measurement time.

Temperature and humidity Extract air + 20 °C, 30 % relative humidity

Outdoor air 1 - 7 °C

Outdoor air 2 −15 °C

NOTE 2 It is advised to consider special applications, e.g. high exhaust air humidity, other duty points.

Sampling for performance testing should be done during a number of complete defrosting cycles. Total testing time should include at least 3 cycles and the minimum test time should be 6 h. For non-cyclic defrosting, sampling should be done under steady-state conditions. This condition is reached when

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temperatures are stable and the heat exchanger exhaust pressure difference, Δ_{p1}, does not vary more than 5 % from the mean value during test.

Functioning of the defrosting equipment should be controlled at the same duty points. When testing a cyclic defrosting arrangement, the variation of pressure difference Δp1 between different cycles should not exceed 5 %. Non-cyclic systems can be considered to meet the requirements if they become stable under operating conditions.

A.3.3 Test procedures

Where appropriate and not otherwise stated, EN 308 should apply. In addition to the applicable requirements in EN 308, the test should be in accordance with the procedures given in A.3.4 and A.3.5.

A.3.4 Testing of defrosting heat factor

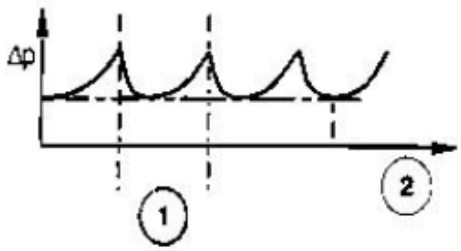
The mean value of the temperatures in both sections, t_{11} and t_{21}, should be adjusted to within 1 °C of the given temperature in A.3.2 during the test. The maximum deviation should be within 1,5 °C from the mean value during the test.

Any energy input for defrosting should be taken into account for calculating the defrosting factors. The temperatures of the air flows, defrosting energy and pressure difference should be measured continuously during the test. The sampling interval should not exceed 60 s.

A.3.5 Total measuring time

For non-cyclic defrosting the measuring time after reaching steady-state condition should be 30 min.

For cyclic defrosting the total measuring time should be a minimum of three cycles, according to Figure A.2.



Key

1 measuring time

2 time s

Figure A.2 — Measuring time for cyclic defrosting

A.4 Test report

A.4.1 Heat recovery device

A description should be made of the defrosting arrangement. Any adjustment of parameters to control defrosting such as time and temperatures should be clearly stated in the report.

A.4.2 Defrosting heat factor

The following parameters should be presented:

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— nominal value of the parameters at the beginning of the test, the mean value of the parameters, and diagrams showing the parameters as a function of time during the test:

q_{m1}	q_{m2}	t_{11}	t_{21}	ε_D	$\dot{Q}_{defr}$	Δp_1
$kg \cdot s^{-1}$	$kg \cdot s^{-1}$	°C	°C	%	W	Pa

— total measuring time;

— cycle time.

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Annex B
(informative)

Air handling units – Heat recovery – Characteristics

B.1 Efficiency of the heat recovery

The thermal quality of a heat recovery system (HRS) is determined considerably by the ratio of thermal changes (η_t) (temperature efficiency). With a possible humidity transmission the efficiency of a heat recovery can be described also by the enthalpy transmission efficiency or the combination of changes of temperature and humidity (humidity efficiency).

The temperature efficiency indicates the relationship of the possible temperatures exchange of a HRS to the maximally possible exchange, thus to the temperature potential between outside and exhaust air. It results from the heat balances:

η_t = use of the HRS / potential of the HRS

$$\eta_t = \frac{\dot{Q}_{HRS}}{\dot{Q}_P} \tag{B.1}$$

where

$\dot{Q}_{HRS}$ is the capacity of the HRS, [kW];

$\dot{Q}_P$ is the maximally possible capacity due to the temperature potential, [kW];

with

$$\dot{Q}_{HRS} = q_{m2} \cdot c_{pA} \cdot (t_{22} - t_{21}) \tag{B.2}$$

or

$$\dot{Q}_{HRS} = q_{m2} \cdot (h_{22} - h_{21}) \tag{B.3}$$

where

q_m is the mass flow of the air, [kg · s^{-1}];

c_{pA} is the specific thermal capacity, [kJ · kg^{-1} · K^{-1}];

t is the temperature of the air, [°C];

h is the enthalpy of the air, [kJ · kg^{-1}].

The maximally possible capacity is formed by the temperature potential, thus the temperature difference between extract air (t_{11}) and outside air (t_{21}), (see Figure B.1).

Thus follows:

$$\eta_t = \frac{\dot{Q}_{HRS}}{\dot{Q}_P} = \frac{q_{m2} \cdot c_{pA} \cdot (t_{22} - t_{21})}{q_{m2} \cdot c_{pA} \cdot (t_{11} - t_{21})} \tag{B.4}$$

And the temperature efficiency:

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$$\eta_t = \frac{(t_{22} - t_{21})}{(t_{11} - t_{21})} \tag{B.5}$$

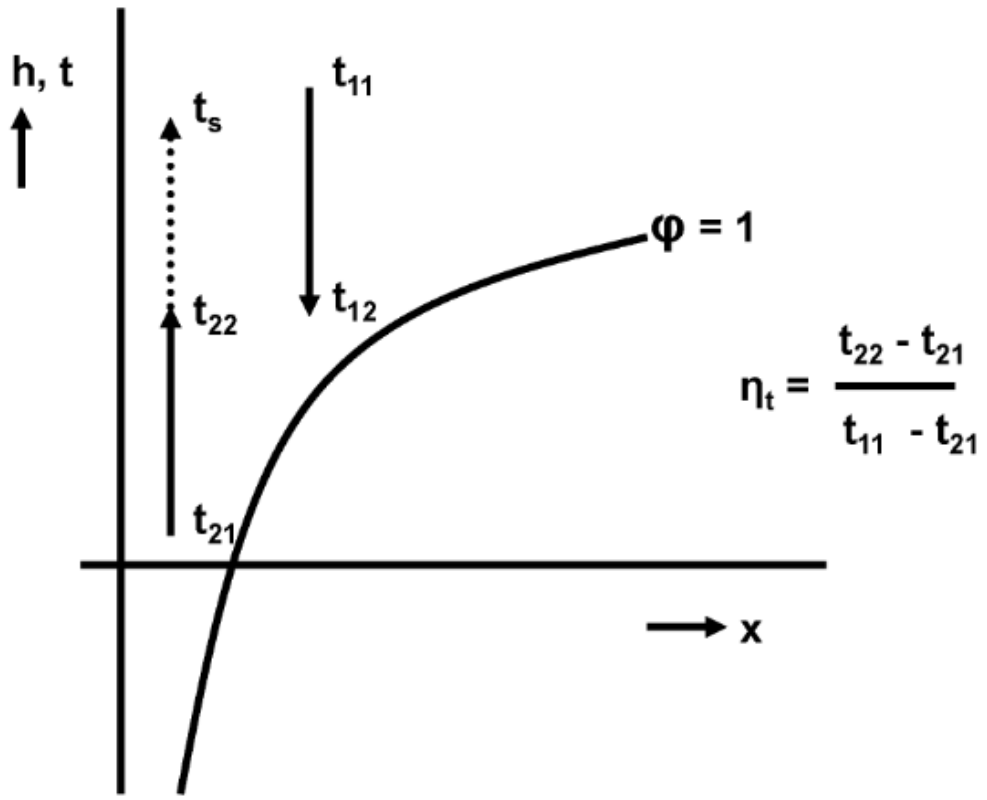

Figure B.1 — Heat recovery in hx-diagram (winter operation, supply air to be heated after heat recovery to temperature t_s)

In the case of a possible humidity transmission the efficiency of humidity (η_x) results similarly out:

$$\eta_x = \frac{(x_{22} - x_{21})}{(x_{11} - x_{21})} \tag{B.6}$$

where

x is the absolute humidity of the air, [g · kg^{-1}].

Here it is to be noted that the efficiency of the humidity transmission is not constant contrary to the ratio of temperature transmission and strongly depends on the humidity difference between the two air flows. This potential κ is calculated by:

$$\kappa = x_{11} - x_{2s} \tag{B.7}$$

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where

x_{2s} is the saturation humidity of the cold air flow x_{21}.

With sorptive heat exchangers the potential depends additionally on the temperature difference of the two air flows.

Summarized thereby the enthalpy efficiency can be calculated with:

$$\eta_h = \frac{(h_{22} - h_{21})}{(h_{11} - h_{21})} \tag{B.8}$$

with

$$h = c_{pA} \cdot t + x \cdot (c_{pS} \cdot t + r_0) \tag{B.9}$$

where

c_{pA} is the specific thermal capacity of air, [kJ · (kg · K)$^{-1}$];

t is the temperature, [°C];

c_{pS} is the specific thermal capacity of water vapour, [kJ · (kg · K)$^{-1}$];

r_0 is the heat for vaporization of water, [kJ· kg^{-1}].

Under dry conditions with $\Delta x = 0$ the enthalpy is calculated with:

$$h = c_{pA} \cdot t \tag{B.10}$$

Under this condition the temperature efficiency ηt is equal to the enthalpy efficiency η_{th}. The efficiencies are defined in accordance to EN 308 only on the supply air side, in order to avoid mistakes. To definite the efficiencies related to the exhaust air side is possible accordingly.

B.2 Evaluation

Condensation on the exhaust side can be excluded in evaluation due to relatively short time of condensation conditions. Therefore the definition of efficiencies under dry conditions is indispensable.

Under condensation conditions on the exhaust side the efficiency rises clearly by the improved heat transmission and the more favourable temperature differences under the latent enthalpy portion. But due to the small frequency (in hours) of these conditions this has hardly an influence on the economic values of the HRS.

In addition to the thermal efficiency the HRS is determined also by the pressure losses.

B.3 Evaluation of auxiliary energies

The pressure losses of the HRS determine auxiliary energies and work, which are necessary to keep the HRS running. These auxiliary energies essentially determined by the electric drives (fans and other energy consuming equipment, e.g. pumps). The necessary power input is calculated thereby:

$$P_{el} = q_v \cdot \Delta p_{HRS} \cdot \frac{1}{\eta_D} + P_{el,aux} \tag{B.11}$$

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where

P_{el} is the electric power input, [W];

q_v is the air or media flow, [$m^3 \cdot s^{-1}$];

Δp_{HRS} is the sum of pressure losses, supply and exhaust, of the HRS, [Pa];

η_D is the system efficiency of the drive system (e.g. fans), [./.];

$P_{el,aux.}$ is the auxiliary electric power input (e.g. pumps), [W].

The ratio between electrical power input and thermal capacities can be described also by a coefficient of performance (ε):

$$\varepsilon = \frac{\dot{Q}_{HRS}}{P_{el}} \tag{B.12}$$

B.4 Further characteristics

Further characteristics of HRS,either on the capacities of the HRS or on the energies or work, which are calculated on yearly basis, can also be defined. However, these are not dealt with in this standard.

B.5 Efficiency

The efficiency of the HRS can be defined by the thermal and the electrical capacities. If no auxiliary energy would be needed to run the HRS, the temperature efficiency would be equal to the thermal efficiency of the HRS. The efficiency represents a combined value from the temperature efficiency and the COP (ε):

η_e = use of the HRS / potential of the HRS

$\eta_e = (\dot{Q}_{HRS} - P_{el}) / \dot{Q}_P$

and

$\eta_e = (1 - P_{el} / \dot{Q}_{HRS}) / (\dot{Q}_P / \dot{Q}_{HRS})$

$\eta_e = (1 - 1 / \varepsilon) / (1 / \eta_t)$

$\eta_e = \eta_t \cdot (1 - 1 / \varepsilon)$

$\eta_e = \eta_t \cdot (1 - P_{el} / \dot{Q}_{HRS})$

The representatively annual efficiency should be based on yearly energies (work), related to yearly energy calculations. The thermal capacities depend very strongly on the outside temperature, while the electrical capacities of auxiliary energies are relatively constant over the year. This method of yearly energy views can be transferred also to further characteristic values of the HRS.

B.6 View of yearly energy

Exactly the same way characteristic values from the total work can be determined after the calculation of the yearly work values. This has the advantage of average characteristic values of the HRS. This gives a better possibility of the evaluation as capacity-referred characteristic values. But these values have a very high influence on the basic conditions of the calculation.

Impact of frost protection if required shall be included.

Yearly work COP:

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$$\varepsilon = \frac{W_{\mathrm{HRS}}}{W_{\mathrm{el}}} \tag{B.13}$$

with

$W = \Sigma\,(Q \cdot t)$ [kWh]

yearly efficiency:

η_{ea} = Use of the HRS / potential of the HRS

$\eta_{\mathrm{ea}} = (1 - W_{\mathrm{el}} / W_{\mathrm{HRS}}) / (W_{\mathrm{P}} / W_{\mathrm{HRS}})$

$\eta_{\mathrm{ea}} = \eta_{\mathrm{ta}} \times (1 - 1 / \varepsilon_{\mathrm{a}})$

yearly efficiency in primary energy:

$\eta_{\mathrm{ea}} = (1 - f \cdot W_{\mathrm{el}} / W_{\mathrm{HRS}}) / (W_{\mathrm{P}} / W_{\mathrm{HRS}})$

$\eta_{\mathrm{ea}} = \eta_{\mathrm{t}} \times (1 - f / \varepsilon_{\mathrm{a}})$

with

f = primary energy factor (e.g. $f = 2$).

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Annex C
(informative)

Static pressure measurement

C.1 Pressure measurement inside a unit

To measure pressure inside the unit it is recommended either using a pressure measurement box or ring lines under the following conditions:

For both methods:

— placed at a fluidically quiet location;

— at a distance from stagnation regions; and

— the fan shall not blow directly at the box/line, and if the fan blows along a surface, for example, at the bottom of the unit, the box/line cannot be placed at this surface.

For pressure relief box only:

— maximum size: L = 80 mm, W = 80 mm, H = 80 mm;

— prepare with only one hole (0,5 to 3 mm) at the bottom (backside) of the box (centre);

— the backside of the box shall be equipped with spacers (distance buds) that secure a distance between the box and the casing of approx. 1-2 mm; and

— located on a plane surface.

For ring line only:

— shall be placed along all four inside surfaces of the unit at each measuring plane;

— tubes in maximum ø10 mm ± 1 mm; and

— shall contain a minimum of four holes per of side (surface) of maximum 1,5 mm ± 0,2 mm pr. side.

Measurements of the static pressure, before and after non-ventilation components, that cannot be dismounted and removed and for which it is impossible to use the four connected pressure relief boxes (electric box) or the ring-line method (see test setup) then measuring with four connected pressure taps in flush with the internal casing is acceptable.

If the non-ventilation components are placed close to each other serial or in a group, the pressure difference can be measured for the entire group as follows: Static pressure after group - static pressure before group.

If the non-ventilation components are placed separately, the pressure difference shall be measured for each component and summarized.

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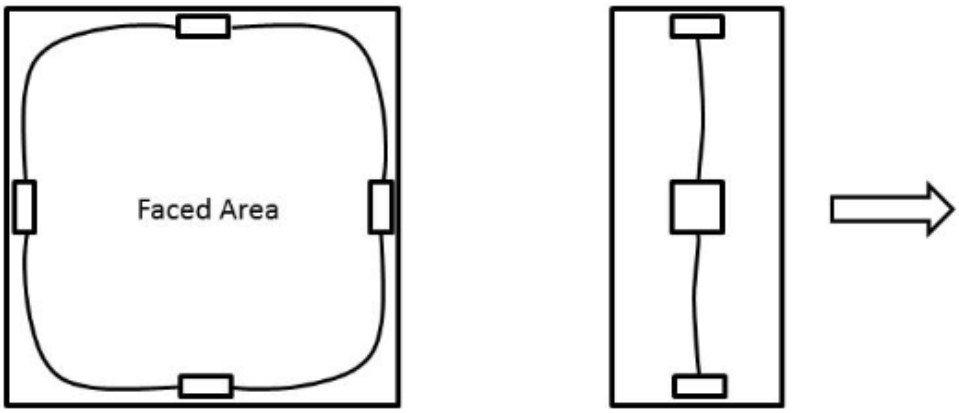

Figure C.1 — Measurement of static pressure in a duct with pressure measurement box

C.2 Pressure measurements in duct-connections outside the unit

Pressure measured external according to ISO 5801. If the duct connections are rectangular, and the velocity is below 10 $m \cdot s^{-1}$ in the ducts connections at the out/inlet of the VU, and the duct connections have a length and width above 500 mm, the external pressure can be measured in the transitions mounted on the unit inlets and outlets.

Four connected (with tubes) pressure relief boxes or a ring-line shall be placed as showed in Figure C.1 in the transitions mounted at the in-and outlets on the unit. The transition has to be made with a straight duct before the start angle for placing the pressure measurement boxes /ring line.

The angle of the transitions may not exceed 15°. The length of the straight duct shall be at least half times the maximum transverse dimension before and after the pressure measuring point.

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Annex ZA
(informative)

Relationship between this European Standard and the ecodesign requirements of Commission Regulation (EU) No 1253/2014 aimed to be covered

This European Standard has been prepared under a Commission's standardization request M/537/C (2015) 8325 final to provide one voluntary means of conforming to the ecodesign requirements of Commission Regulation (EU) No 1253/2014 of 7 July 2014 implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to ecodesign requirements for ventilation units [L 337/8].

Once this standard is cited in the Official Journal of the European Union under that Regulation, compliance with the normative clauses of this standard given in Table ZA.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding ecodesign requirements of that Regulation and associated EFTA regulations.

Table ZA.1 — Correspondence between this European Standard and Commission Regulation (EU) No 1253/2014] of 7 July 2014 implementing Directive 2009/125/EC of the European Parliament and of the Council with regard to ecodesign requirements for ventilation units [L 337/8] and Commission's standardization request M/537/C (2015) 8325 final

Ecodesign Requirements of Regulation EU No 1253/2014 [L 337/8]	Clause(s)/subclause(s) of this EN	Remarks/Notes
3.2	6.5 Heat recovery sections 6.5.2 Classifications and requirements 5.4.2.1.2 Built-in fan efficiency 5.4.2.2.2 Built-in fan efficiency 5.4 Testing of internal specific fan power of ventilation components	Heat Recovery System Minimum thermal efficiency (η_t for balanced mass flow) Minimum fan efficiency for UVUs ($\eta_{s,fan}$ is compared with η_{vu} limit to show compliance with the ventilation unit regulation) Internal specific fan power
3.4	6.5 Heat recovery sections 6.5.2 Classifications and requirements	Heat Recovery System Minimum thermal efficiency (η_t for balanced mass flow)

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	5.4.2.1.2 Built-in fan efficiency 5.4.2.2.2 Built-in fan efficiency 5.4 Testing of internal specific fan power of ventilation components	Minimum fan efficiency for UVUs ($\eta_{s,fan}$ is compared with η_{vu} limit to show compliance with the ventilation unit regulation) Internal specific fan power
4.2	6.5.2 Classifications and requirements 3.22 nominal air flow 5.4.1.2 Quantities 5.4 Testing of internal specific fan power of ventilation components 6.2 Casing 3.17 nominal external pressure 3.18 internal pressure difference of ventilation components 6.9.2 Filters installed in air handling units 5.3.2.1 Casing radiated noise tests	thermal efficiency of heat recovery nominal NRVU flow rate effective electric power input SFP_{int} face velocity at design flow rate nominal external pressure ($\Delta p_{s,ext}$) internal pressure drop of ventilation components ($\Delta p_{s,int}$) description of visual filter warning for NRVUs the casing sound power level (LWA)

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European Standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the products falling within the scope of this standard.

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Bibliography

[1] EN 305, *Heat exchangers — Definitions of performance of heat exchangers and the general test procedure for establishing performance of all heat exchangers*

[2] EN 10088-2, *Stainless steels — Part 2: Technical delivery conditions for sheet/plate and strip of corrosion resisting steels for general purposes*

[3] EN 13030, *Ventilation for buildings — Terminals — Performance testing of louvres subjected to simulated rain*

[4] IEC 60034-2-1, *Rotating electrical machines — Part 2-1: Standard methods for determining losses and efficiency from tests (excluding machines for traction vehicles)*

[5] ISO 5168, *Measurement of fluid flow — Procedures for the evaluation of uncertainties*

[6] ISO 13348, *Industrial fans — Tolerances, methods of conversion and technical data presentation*